



Quantum-inspired metaheuristic algorithms: comprehensive survey and classification

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Abstract

Metaheuristic algorithms are widely known as efficient solutions for solving problems of optimization. These algorithms supply powerful instruments with significant engineering, industry, and science applications. The Quantum-inspired metaheuristic algorithms were developed by integrating Quantum Computing (QC) concepts into metaheuristic algorithms. The QC-inspired metaheuristic algorithms solve combinatorial and numerical optimization problems to achieve higher-performing results than conventional metaheuristic algorithms. The QC is used more than any other strategy for accelerating convergence, enhancing exploration, and exploitation, significantly influencing metaheuristic algorithms' performance. The QC is a new field of research that includes elements from mathematics, physics, and computing. QC has attracted increasing attention among scientists, technologists, and industrialists. During the current decade, it has provided a research platform for the scientific, technical, and industrial areas. In QC, metaheuristic algorithms can be improved by the parallel processing feature. This feature helps to find the best solutions for optimization problems. The Quantum-inspired metaheuristic algorithms have been used in the optimization fields. In this paper, a review of different usages of QC in metaheuristics has been presented. This review also shows a classification of the Quantum-inspired metaheuristic algorithms in optimization problems and discusses their applications in science and engineering. This review paper's main aims are to give an overview and review the Quantum-inspired metaheuristic algorithms applications.

Keywords Metaheuristics · Optimization · Quantum Computing · Quantum-Inspired · Combinatorial

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1 Introduction

Nowadays, advanced non-linear optimization problems are complex and harder to solve since they are described in high-dimensional spaces where there is insufficient data in certain instances to form an issue mathematically [1, 2]. Mutual heuristic styles cannot offer sophisticated solutions due to such challenges, and therefore there is much gravitation towards non-exact new optimization approaches called “metaheuristic algorithms”. The term metaheuristic was launched by Glover [3] and represented a class of well-defined algorithms for solving complex optimization problems. Metaheuristic algorithms search the problem space out of an unvaried heuristic process and create a suitable solution [4–6]. Metaheuristics are non-gradient-based methods that usually deal with mathematical methods derived from a physical or social rule or a natural event. Metaheuristics are algorithmic frames applied for various optimization problems with only a few changes for adapting to the given problem. Also, metaheuristics have basic features that can be summarized as follows:

- They are used for various issues.
- They are generally approximate to achieve solutions.
- They scout the search area to find an acceptable solution.
- They can fundamentally be defined by secession level.
- They may include a variety of mechanisms to avoid premature convergence.
- A metaheuristic using computational methods is domain-specific knowledge dominated by the higher-level strategy.
- They generally permit a comfortable parallel performance.
- They are expanded from primer local search to advanced learning methods.
- Metaheuristic algorithms use memory and search to find optimal solutions.

Process Optimization is used to identify the valuable variables to reduce or increase their values. Almost all real-world issues have non-linear limits and high computational costs, are non-convex and complicated and have many solution areas [7, 8]. Consequently, it is complex and hard to solve such problems with numerous variables and constraints [9–11]. Secondly, many local optimum solutions do not promise the best using traditional numerical methods [12]. To prevail over these problems, metaheuristic optimization algorithms are implemented to solve such complicated problems [13–15]. The development and implementation of metaheuristics have extraordinary importance for flexibility, reasonable computational time, and simplicity [16].

The main concept behind the design of metaheuristic algorithms is to address complex optimization problems where other optimization methods have failed to be effective [17]. These methods are now distinguished as some of the most practical approaches for solving many real-world problems. There are several advantages to using metaheuristic algorithms for optimization [18, 19].

- Hybridization: They can be combined with more general optimization techniques.
- Extended applicability: They can be applied to every problem that can be formulated as function optimization problems.
- Ease of performance: Especially easier to understand and implement.

- Efficiency and flexibility: Larger problems can be solved faster. They are simple to design and implement and are very flexible compared to others.
- The application of metaheuristics can be justified due to: (I) the intricacy of the internal problem that bans the application of exact techniques and (II) a large number of possible solutions that prevent the use of perfect algorithms [20].

However, there are some problems with the metaheuristic algorithms that should be noted here[21]:

- Usually, optimization proficiency is highly tied to fine element tuning.
- Metaheuristics do not have a fixed mathematical basis when compared to older techniques.
- They cannot prove optimality.
- They cannot minimize the search space.
- Repeatability of the optimization results gained with the same primary condition settings is not imminent.

The population-based metaheuristic algorithms are extensively divided into three classes: the Physic-based, Evolutionary-based, and Swarm-based methods. The Physics-based algorithms are in the first group where search particles can communicate the space's perimeter based on physics rules such as inertia, electromagnetic, and gravitational forces[22, 23]. The general structure of these algorithms differs from other methods because of how the search represents physics rules. The Physics-based optimization algorithms include Simulated Annealing (SA) [24] and Gravitational Search Algorithm (GSA) [25]. The SA is derived from annealing in metallurgy that provides heating and controlled cooling qualities of a material. These properties rely on thermodynamic free energy. GSA is formed based on the rule of gravity and mass interactions. The population solutions work with each other via gravity force, and the mass evaluates their function. Some of such algorithms are Charged System Search (CSS) [26], Small-World Optimization Algorithm (SWOA) [27], and Water Wave Optimization (WWO) [28].

Evolutionary Algorithms (EAs) are in the second sub-category of metaheuristic algorithms. These algorithms are based on the theory of natural selection and biological evolution. These algorithms frequently act well to find optimal solutions due to not knowing the entire fitness landscape. The most common EA is the Genetic Algorithm (GA). GA was first proposed by Holland [29]. Individuals are created from a casual population with evolution transmutation. In each generation, the fitness criterion is computed for each individual. The intersecting and transformation operators are employed on the individuals to make a new and robust population. Thus, the best individuals can create a new population because of their higher possibility through iterations. Differential Evolution (DE) [30] is another Evolutionary-based algorithm that optimizes an issue by keeping a candidate solution and making new candidate solutions by combining the existing ones. It maintains the nominated solution with the best fitness value for optimization issues. In addition to these, some other well-known evolutionary-based algorithms are Genetic Programming (GP) [29] and Social Evolution Algorithm (SEA) [31].

The third main branch in metaheuristics is Swarm-based algorithms based on social creatures' collective behaviour [32]. These methods have usually derived from natural colo-

nies, flocks, and herds. Swarm-based algorithms' objective is to solve problems by modeling the populations of individuals that can self-arrange and interact with each other [33]. Individuals can switch information to share their experiences. The combination of actions and interactions of these simple factors helps increase population performance's power. All metaheuristic algorithms are based on exploring and exploiting a search area [34–36] and keeping a suitable balance between exploration and exploitation. The exploration stage in an algorithm probes different promising areas in a search space, whereas exploitation can look for optimal solutions near the promising regions.

Consequently, these two stages of fair regulation must get near-optimal solutions. In the previous few years, with the fast stimulation of advanced non-linear optimization issues in different science spaces, there has been an instant need for effective and efficient optimization strategies to interact with these increasingly complex optimization issues [37, 38]. Local solutions are critical when looking for optimization problems because only one best solution in the search space will exist.

Nonetheless, the solutions found with higher frequency will be closer to the solution of the objective function. These solutions can be considered the best local solutions, though not the best global ones, considering the whole search space [39]. These solutions may stick to the local optimum because of regional strategies. The convergence speed is another issue because an algorithm that gets away from local solutions does not imperatively converge to the global optimum [40]. Despite the considerable number of recently-extended metaheuristic algorithms, the upcoming question is how more meta-heuristic techniques must be developed.

The best way to increase the power of meta-heuristic algorithms is QC [41, 42]. The function of a metaheuristic algorithm for a typical set of issues does not act to solve other optimization problems due to their different operators. The QC permits researchers to propose efficient metaheuristic algorithms for answering issues in various areas [43, 44]. The hybridization of a metaheuristic technique is vital in finding answers to problems. The performance of nature-inspired optimization algorithms was frequently studied and improved in the previous ten years using QC-based techniques. To solve problems involving large-scale optimization, high dimensional problems, and dynamic optimization, for example, researchers frequently adapt the original algorithm to QC. Some evolutionary processes, like selection, crossover, and mutation for GA, are carried out based on QC.

When making judgments or analyzing systems, optimization is a crucial tool. Finding optimal or nearly optimal solutions among all practical options concerning specific goals is an optimization problem. The objective expressed as a quantitative performance measure of the optimized system, the system's unknown variables, and the constraints of the problem, which specify the relationships between the variables and their permitted values, are all expressed using optimization algorithms to solve optimization problems. Because they can't be addressed optimally in an "acceptable" amount of time, specific optimization tasks are inherently more difficult than others. QC-metaheuristic techniques are frequently used to approach this type of problem.

Hybridization methods can be used for optimal metaheuristic algorithms [45–49]. Generally, a metaheuristic algorithm balances the global and local search methods to improve the solutions' fitness. Nevertheless, the local search is commonly performed by random (i.e., random walk) around the best solutions. Accordingly, a high measuring effort can be applied without effective local search [50, 51]. To entangle the limitations mentioned before, it is

possible to develop hybrid attitudes that take the benefits of each category of algorithms. Therefore, QC combines the global search operation metaheuristics and typical optimization methods to perform an effective local search. The delicate balance between these two ingredients is exceptionally vital to the efficiency and performance of an algorithm. Suppose there are too little exploration and too much exploitation. In that case, it will cause the system to be trapped in local optima, making it very difficult or even impossible to find the global optimum [52, 53]. From another point of view, it may be difficult for the system to converge on the condition of too much exploration but too little exploitation, and therefore it slows down the general search [54]. The primary difference among the evolved metaheuristic algorithms is the approach to acquiring such stability.

Research on metaheuristic algorithms for Combinatorial Optimization Problems (COPs) has recently experienced a considerable shift towards the combination of metaheuristics with different strategies for optimization. Metaheuristic algorithms' performance improves with QC, leads to optimal solutions, and accelerates the convergence rate. The combination of other metaheuristics may be very sizeable, mainly for using quality controls inside primarily population-based strategies. Metaheuristics frequently use QC for refining the solutions generated during the quest technique [54]. It could be attributed to the fact that these nature-stimulated techniques are suitable for exploring the quest space and the identity of areas with high-quality solutions. They typically try to seize an international photo of the quest area at initialisation. Then, they successfully direct the search to different promising regions of the search space at some point in the search process. The strength of QC is locating more valuable solutions within the part of given starting solutions faster. In other words, primarily population-based methods correctly identify the search space's promising areas, wherein quality control can then determine the best answers quickly [54–56].

The QC is a new computing class based on quantum implication ideas and standards, including the superposition of quantum states, entanglement, and intermediation. Quality controls are the art of applying the legal guidelines of quantum mechanics to computer science. The idea of QC changed into physicists' proposed form in the early 1980s [57]. The QC is more vigorous than classical computing in solving computational complexity theories. The smallest unit of quantum information is a quantum bit (Q-bit). Quantum computers use a series of quantum operations on Q-bits. Every Q-bit can be formulated in states "1", "0", or an aggregate of both states on the equal time, called superposition. A Q-bit is the minimum amount of technique-capable facts in QC: a two-dimensional quantum-mechanical device encoding the classical bits of records 0 and 1 in their necessary states: $|0\rangle$ and $|1\rangle$. This system is likely to exist in a superposition of states $|0\rangle$ and $|1\rangle$. It is the primary acute property of quantum structures: they may simultaneously be in all the system's states. This feature allows quantum computer systems to carry out parallel computations on a large scale. An ordinary computer can most effectively be in any such 2^n states simultaneously. A Q-bit is indicated as a pair of numbers (α, β) , wherein the values $|\alpha|^2$ and $|\beta|^2$ constitute the possibility of finding the Q-bit within the states "0" and "1", respectively [57–59].

This paper focuses on quantum-inspired metaheuristic algorithms. In the past, many solutions have been proposed to solve optimization problems based on quantum-inspired metaheuristic algorithms. As discussed in the introduction, complex optimization issues are a more challenging task. Analyzing the existing solutions can help understand the optimization problem and identify other complex issues.

This paper fills the gap and makes four essential contributions:

- Although some works of quantum-inspired metaheuristic algorithms exist, they cannot provide a comprehensive view. The current paper reviews papers based on advantages, disadvantages, goals and applications.
- In this paper, all meta-heuristic algorithms have been analyzed based on QC.
- It explained various classifications and different applications of QC. The Quantum-inspired meta-heuristic algorithms are classified as Hybrid, Binary, Chaotic, and Multi-objective strategies.
- This paper presents a complete review of the quantum-inspired meta-heuristic algorithms proposed in papers related to different optimization fields and suggests potential future directions.

This paper is organized as follows. In Sect. 2, the fundamentals of QC have been reviewed. In Sect. 3, the QC methods (QC-Metaheuristics, Variants of QC, and QC-Machine Learning) have been discussed. In Sect. 4, QC applications have been discussed. Finally, the conclusion and suggestions for future works have been presented in Sect. 5.

2 Quantum computing

Feynman [57] first introduced the idea of QC. According to his commentary, classical computer systems cannot efficiently simulate quantum-mechanical structures, and other methods must be designed for that purpose [41]. Exponentially cumbersome quantities of facts require a quantum gadget to be epitomized properly. The algorithms executed in classical computers possess sub-exponential areas and time complexity. Classical computer systems cannot be used to simulate quantum gadgets under these circumstances. On the opposite, quantum computer systems are constructed to operate the non-classical functions of quantum systems. This machine exploits the houses above to exponentially more significant quantities of records in polynomial time.

In reality, the computational mechanism for quantum computer systems and their classical counterparts are immeasurably distinctive. The primary extraordinary function of a quantum gadget is called the superposition of quantum mechanics. A quantum system essentially handles a range of viable states at a time. In this machine, a quantum register is configured so that each possible combination of 0 and 1 s simultaneously exists. At the same time, in classical devices, only a single item may be saved in a computer's memory at any stipulated time. A quantum set of rules accepts n quantity of classical bits as input, works on them, and produces a superposition of 2^n possible states as output. After that, those exponentially massive statistics are manipulated through a dimension technique to gain the last quantum results. The symptoms are evaluated with the opposite possibility distribution to acquire the n output bits in a while. A suitable and simple chain quantum operation is completed to govern all superposed exponential states in the middle section. A quantum system is probabilistic. While a quantum device is measured with a certain probability, resulting in a single quantum state. In QC, discovering the correct state can become a considerable solution.

In the quantum system of classical computers, data are represented as zero or 1; however, quantum computers have the particular capability to symbolize information as a superposition of 0 and 1, i.e., each cost is between 0 and 1. This is known as a quantum bit (Q-bit).

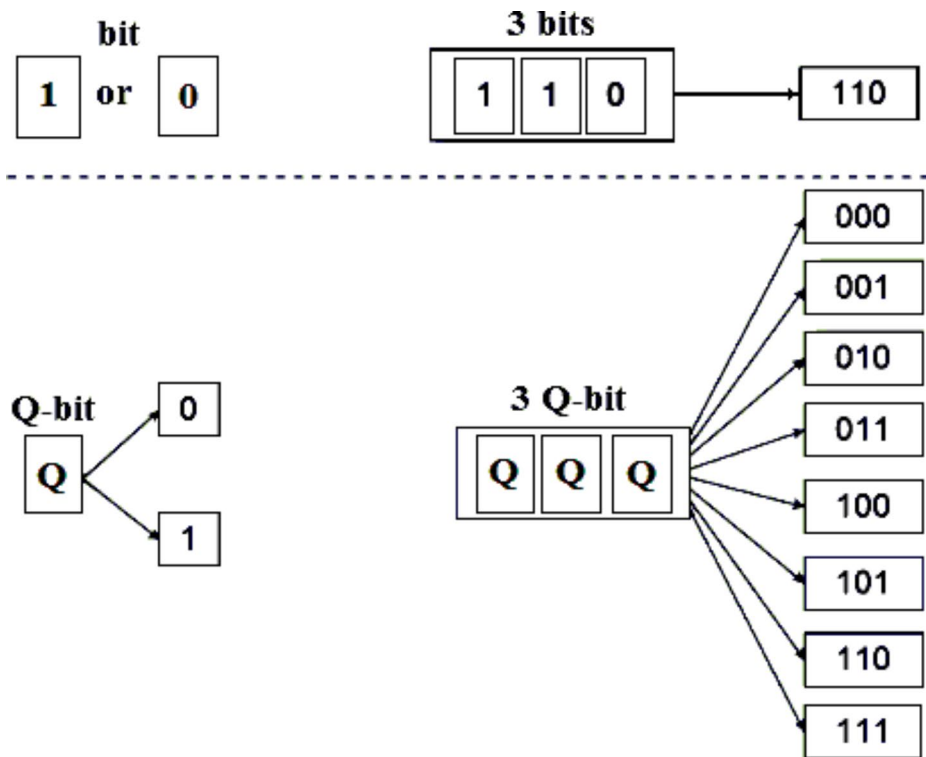


Fig. 1 The difference between a bit and Q-bit [41]

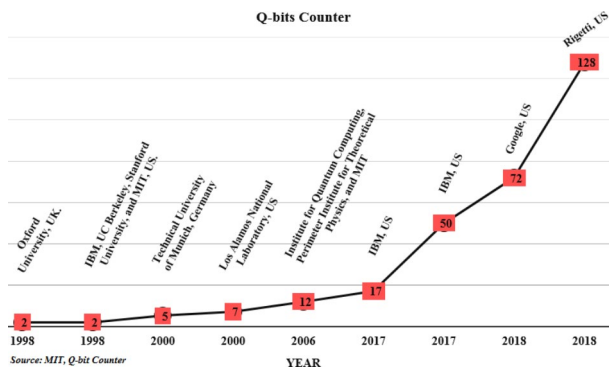
Unlike classical computers that process facts sequentially, a quantum computer can carry out computational processes concurrently, referred to as quantum parallelism. In principle, parallel quantum computer processing may be hundreds, if not thousands, faster than the modern classical computers [41].

Contrarily, in conventional computers that use electricity, the QC saves and processes data based on atoms and primary particles' properties. The classical laptop data is stored in bits with one of the two values, 0 or 1. At the QC, the records are stored in qubits. Each Q-bits can simultaneously deliver two states, 0 and 1 (not one 0 or 1). While appearing as an operation with this qubit (register), the operation may be performed simultaneously and parallel with all feasible combinations with values 0 and 1. With a 3-Q-bit register, each Q-bit can simultaneously be given two states (i.e., 0 and 1). While performing one operation with these three Q-bits registers, the operation may be achieved concurrently (in parallel) with all 2^3 viable mixtures (values) from 000 to 111. This explains the parallelism of quantum computation. Figure (1) indicates the distinction between bit and Q-bit.

Two distinct states characterize both classical bits and quantum bits. The difference is that classical bits can best be in a single state, while a qubit may be in a superposition of an aggregate of the two.

A classical computer takes information, usually as a list of 1s and 0s, i.e., bits (together with electrical indicators at distinct voltage tiers), and uses digital circuitry in the system. It performs a fixed number of pre-decided calculations, which may be broken down into the

Fig. 2 The number of Q-bits studied by year and organization [60]



so-called “gate operations” wherein a few bits’ kingdoms are modified primarily based on other bits’ acknowledged value. The QC outputs the result as a string of particular 1s and 0s. A QC could take information encoded on a quantum state, then carry out the pre-decided “gate operations” in line with quantum mechanics laws, and produce a brand-new quantum state, which may be measured to determine the outcome of the computation. Figure (2) shows the variety of Q-bits studied according to the year and organization between 1998 and 2018.

In 1998, researchers at Oxford University claimed they had made a step forward with computing information using two Q-bits. The rate of evolution was so high in 2017 that IBM increased its Q-bit calculation capacity to 50. In 2018, Google claimed data processing with seventy-two Q-bits. In August, *Rigetti* Computing offered plans for a 128 Q-bits quantum chip [60].

In QC, the Q-bit can be considered a unit of records like a classical bit. In the case of classical bits, an individual binary bit can preserve the value 0 or 1. However, an individual qubit can simultaneously hold the price zero and 1, mentioned as the quantum superposition state [41]. The advantage of using qubits is the rapid execution due to qubits’ coherent nature as they may simultaneously perform and control multiple procedures. The superposition of the primary states in QC may be explained in line with Eq. (1) [41, 61].

$$|\Psi\rangle = \sum_{i=1}^n C_i |V_i\rangle \quad (1)$$

$$|\Psi\rangle = C_1 |V_1\rangle + C_2 |V_2\rangle + \dots + C_n |V_n\rangle \quad (2)$$

In Eq. (1) V_i indicates the i^{th} states and $C_i \in \mathbb{C}$. In the case of a two-state Q-bit, Eq. (2) can be rewritten as $|\Psi\rangle = C_1|0\rangle + C_2|1\rangle$. The state $|0\rangle$ is diagnosed as the “ground state,” and the state $|1\rangle$ is known as the “excited state”; C_i is a complex number, and the ideal situation for quantum orthogonality is represented as follows:

$$\sum_{i=1}^n C_i^2 = 1 \quad (3)$$

Dependence and inconsistency are two striking features of QC. In Eq. (2), $|\Psi\rangle$ defines a linear superposition of the primary states and can be defined according to coherence. An

emphatic demolition of the above-mentioned linear superposition is referred to as inconsistency. A quantum mechanical event, namely the quantum involvement, establishes a unique communication between the existing quantum systems by which the entangled qubit states can speed up so that the performance capability may augment at a tremendous rate. Some magnificent quantum gates are NOT gate, C-NOT gate, rotation gate, Hadamard gate, Pauli-X gate, Pauli-Y gate, Pauli-Z gate, Toffoli gate, controlled phase shift gate, Fredkin gate, etc. A rotation gate is accountable for updating the i th qubit value of (α_i, β_i) , and can be represented as following [41]:

$$\begin{pmatrix} \alpha'_i \\ \beta'_i \end{pmatrix} = \begin{pmatrix} \cos\theta_i & -\sin\theta_i \\ \sin\theta_i & \cos\theta_i \end{pmatrix} \begin{pmatrix} \alpha_i \\ \beta_i \end{pmatrix} \quad (4)$$

As a result of this, for each Q-bit, θ_i illustrates the rotation angle. For a particular problem, the rotation angle is chosen accordingly.

3 QC methods

Metaheuristics are nature-inspired algorithmic systems typically non-deterministic and stochastic and designed to solve unique optimization problems, particularly in complex mathematics. In terms of the computer technological know-how, metaheuristics may have been coined because of the computational strategies, which acclimatize an iterative approach to discover the best candidate solution of given trouble regarding a particular measure of exceptional. A few or no changes may be required alongside the basic framework of metaheuristics in solving a specific problem. Metaheuristic algorithms are a productive alternative to the vast classical processes [62]. On the other hand, QC is an embryonic subject where quantum mechanics' capabilities are used to hoard and process information to resolve computational problems significantly. Over the past few years, researchers have proved that algorithms' new tastes may represent the apposite quality options to cope with various optimization issues. This approach performs the complicated correlation between metaheuristic algorithms and QC. Generally, a potentially decreased computational time is determined to execute quantum stimulated metaheuristic algorithms than their classical models. Moreover, this algorithm maintains a suitable balance between exploration and exploitation.

The primary purpose of QC in metaheuristic algorithms is to optimize the initial population, the operator, and the exploration and exploitation steps. The metaheuristic algorithms include iteration and searching optimization points. These algorithms should use hybrid operators to reach the optimal solution and reduce time. QC combines operators to improve and find the best solutions. The modern idea of conjointly running with QC and distinctive metaheuristic methodologies recommended a unique algorithmic paradigm that is generally tagged as a "Quantum-inspired metaheuristic algorithm". These modern strategies exploit the quantum principles in fundamental metaheuristic frameworks. The opposite synergies between every part are essential for designing such methodologies. This brilliant hybridization method makes these algorithms more helpful in performance and robustness than different comparable algorithms.

Figure (3) categorises the quantum-inspired metaheuristic algorithms, including the algorithms, variants, and quantum machine learning classes. Metaheuristic algorithms are

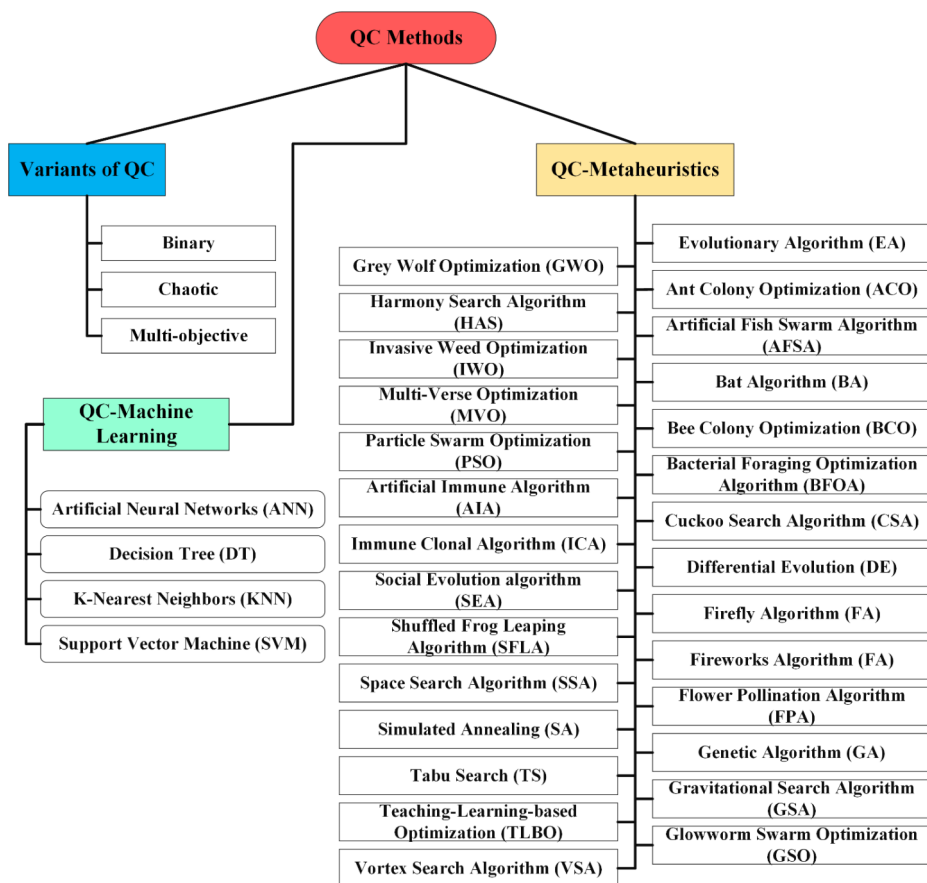


Fig. 3 Classification of QC Methods

within the algorithms class, while the Binary, Chaotic, and Multi-objective strategies are invariants.

Table (1) shows the number of papers related to the Quantum-Inspired metaheuristic algorithms. The EA, ACO, GA, and PSO algorithms have the widest variety of quantum-associated papers. Additionally, the most significant number of posted Quantum-Inspired metaheuristic algorithms belongs to Sciondirect and Springer publishers.

Table (1): The number of papers of Quantum-Inspired metaheuristic algorithms.

Algorithms	Number of Papers	Publisher				
		IEEE	Sciondirect	Springer	Hindawi	Other Journals
Evolutionary Algorithm (EA)	30	√	√	√	-	-
Ant Colony Optimization (ACO)	11	√	√	√	-	-
Artificial Fish Swarm Algorithm (AFSA)	2	√	-	-	√	-
Bat Algorithm (BA)	2	-	-	√	-	-
Bee Colony Optimization (BCO)	3	-	-	√	-	-

Algorithms	Number of Papers	Publisher				
		IEEE	Sciencedirect	Springer	Hindawi	Other Journals
Bacterial Foraging Optimization Algorithm (BFOA)	4	√	√	√	-	√
Cuckoo Search Algorithm (CSA)	3	-	√	√	-	-
Differential Evolution (DE)	7	-	√	√	-	-
Firefly Algorithm (FA)	3	-	√	-	√	-
Fireworks Algorithm (FA)	2	√	-	-	-	√
Flower Pollination Algorithm (FPA)	3	√	-	-	-	√
Genetic Algorithm (GA)	32	√	√	√	√	√
Gravitational Search Algorithm (GSA)	4	-	√	-	-	-
Glowworm Swarm Optimization (GSO)	1	-	-	-	-	√
Grey Wolf Optimization (GWO)	1	-	-	√	-	-
Harmony Search Algorithm (HAS)	2	-	√	-	-	-
Invasive Weed Optimization (IWO)	1	√	-	-	-	-
Multi-Verse Optimization (MVO)	1	-	-	√	-	-
Particle Swarm Optimization (PSO)	28	√	√	√	√	√
Artificial Immune Algorithm (AIA)	3	-	√	-	-	-
Immune Clonal Algorithm (ICA)	5	√	√	√	-	-
Social Evolution algorithm (SEA)	1	-	√	-	-	-
Shuffled Frog Leaping Algorithm (SFLA)	2	-	√	√	-	-
Space Search Algorithm (SSA)	1	-	√	-	-	-
Simulated Annealing (SA)	2	-	-	√	-	-
Tabu Search (TS)	4	√	√	-	-	√
Teaching-Learning-based Optimization (TLBO)	2	-	-	√	-	-
Vortex Search Algorithm (VSA)	1	-	-	√	-	-

Figure (4) shows the number of published Quantum-inspired metaheuristic algorithms per year. The published papers are from reference databases (IEEE, Scencedirect, Springer, Hindawi, and other journals). Using this method, 298 papers were found. Figure (4) suggests the distribution of 298 published papers from 1996 until June 2022. According to Figs. (4), 2021 had the highest frequency of published papers. Figure (5) illustrates the classification of the research via numerous publishers. Figure (6) shows the type of contribution from various publishers.

Figure (5) shows a broad overview of the data obtained from Springer, ScienceDirect, IEEE Xplore, etc. The search strategy implemented uses a combination of keywords to shortlist publications appropriate to this study. Other public databases include Google Scholar, Research Gate, Crossref, etc. This indicates that QC has been emphasized in sci-

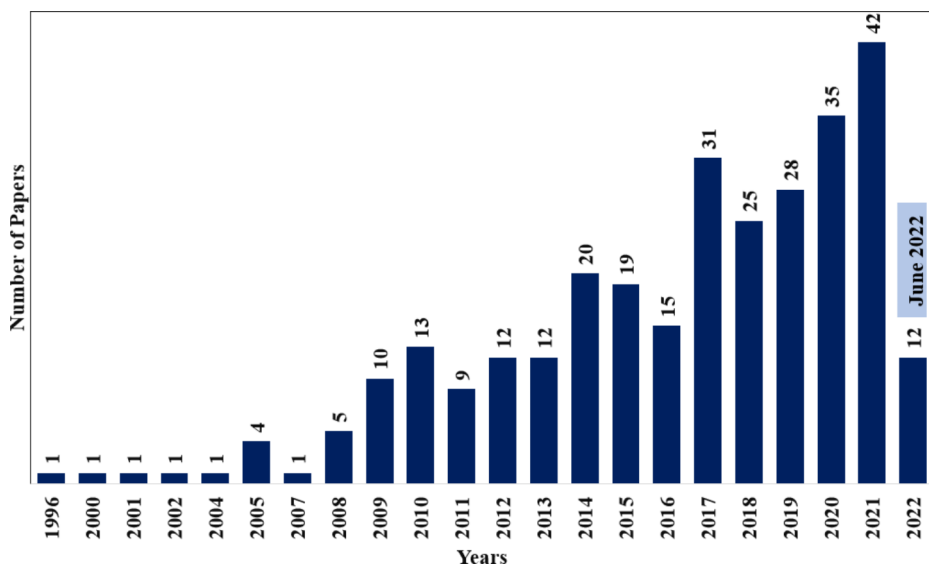


Fig. 4 Distribution of published papers of quantum-inspired metaheuristics by year of publication

entific and metaheuristics research as one of the most distinctive and popular optimization methods with wide application diversity and high performance.

Figure (5) shows that Scienedirect and Springer published 39% and 32% of the papers, respectively. 15% of the papers have been published in IEEE, 6% in Hindawi, and 8% in other Journals. As shown in Fig. (6), journals and conferences posted 65% and 31% of the papers, respectively, while 4% of the papers have been published in the e-book format.

3.1 QC- metaheuristics

3.1.1 Evolutionary algorithm (EA)

Quantum-inspired computational intelligence can be defined as optimization strategies manipulated by quantum bits. These algorithms were created based on the interaction between quality control and evolutionary optimization programming strategies. The Quantum-stimulated techniques can be classified into three predominant groups: the evolutionary-designed quantum algorithms (EDQAs), Quantum EAs (QEAs), and the quantum-inspired EAs (QIEAs). QEA is a probabilistic set of rules exploiting quantum computation's electricity to boost genetic tactics. The primary distinction between a bit and a Q-bit is that the Q-bit can be in a superposition of its allowed states with given possibilities for every state. This could be represented mathematically via a linear mixture of states, usually called the superposition state and represented according to Eq. (5) [63].

$$|\phi\rangle = \alpha |0\rangle + \beta |1\rangle \quad (5)$$

α and β signify complex numbers. This means that the kingdom of a Q-bit is a vector in a two-dimensional complex vector space. The particular states $|0\rangle$ and $|1\rangle$ are the computational

■ IEEE ■ Sciencedirect ■ Springer ■ Hindawi ■ Other Journals

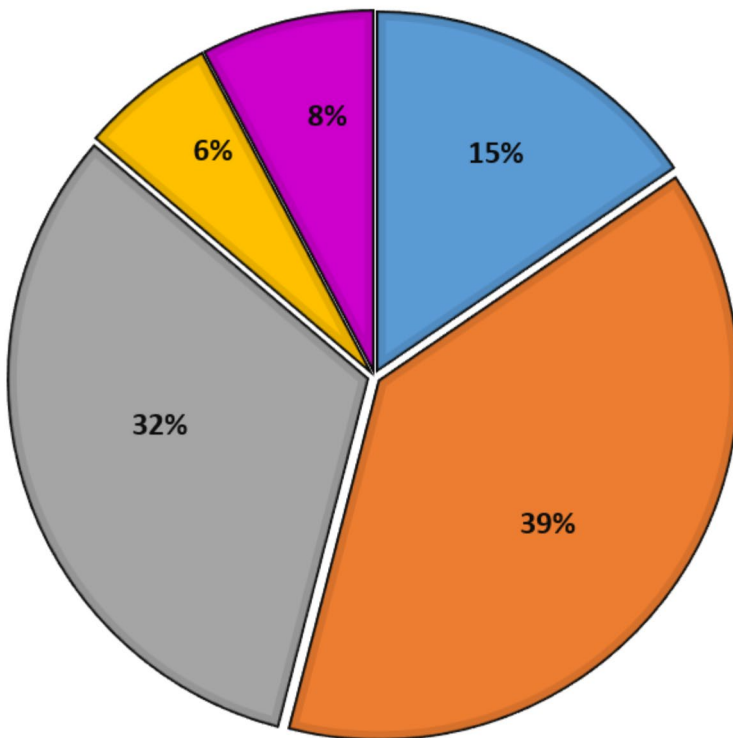


Fig. 5 Percentage of published papers based on different publishers

basis states and shape an orthonormal foundation for this vector area. In $|\alpha|^2 + |\beta|^2 = 1$, α and β are complex numbers that explain the chance amplitudes of the corresponding quantum state. A Q-bit representation of them-(Q-bits) chromosome can be defined as the following [63]:

$$\begin{bmatrix} \alpha_1 & | & \alpha_2 & | & \dots & | & \alpha_m \\ \beta_1 & | & \beta_2 & | & \dots & | & \beta_m \end{bmatrix} \quad (6)$$

where, $|\alpha_i|^2 + |\beta_i|^2 = 1, i = 1, 2, \dots, m$. In this manner, them (Q-bits) chromosomes can simultaneously constitute the facts of 2^m binary states. Because it can represent a linear superposition of many states, QEA has a higher range function than the classical methods. Classical computer circuits encompass wires and logic gates. The wires carry information around the circuit, while the logic gates manipulate the statistics. For the update technique of QEA, the appropriate quantum gate $U(\theta)$ is usually followed in compliance with practical optimization issues. The overall expression of $U(\theta)$ is proven in step with Eq. (7) [63].

$$U(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \quad (7)$$

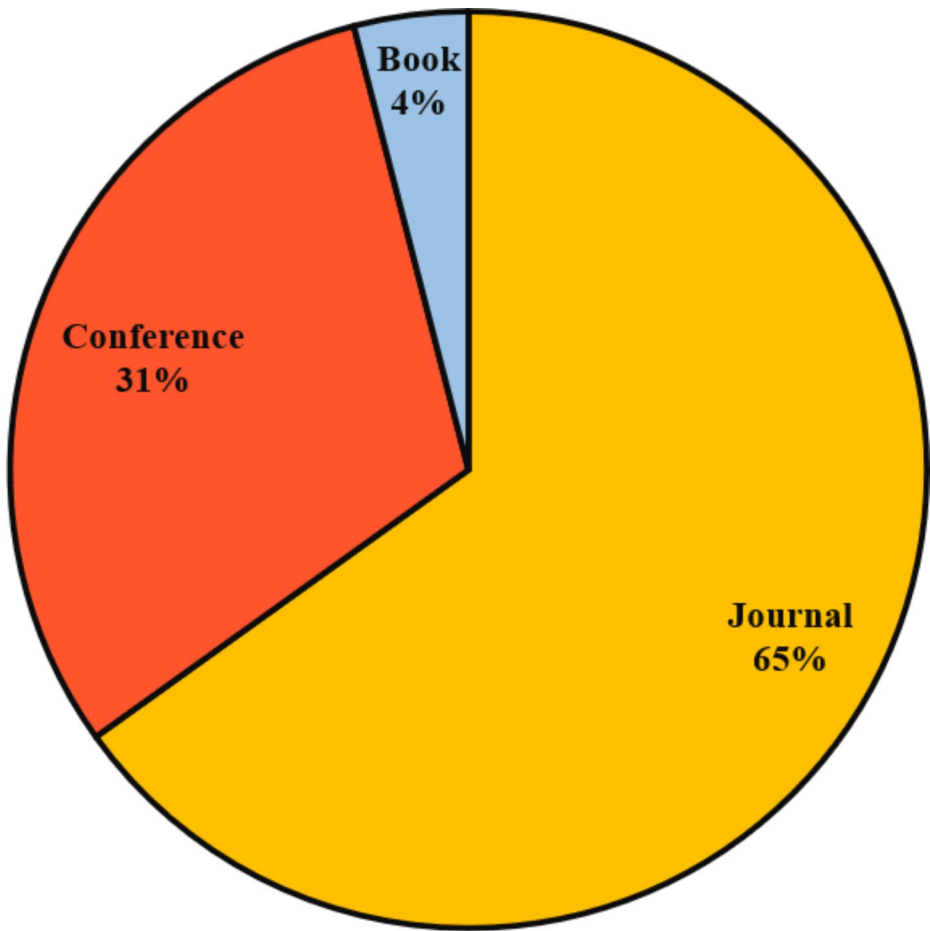


Fig. 6 Percentage of published papers based on the type of contribution

In Algorithm (1), the process of QEA is proven [63].

```

1. Procedure QEA
2. begin
3.  $t \leftarrow 0$ 
4. Initialize  $Q(t)$ 
5. make  $P(t)$  by observing the states of  $Q(t)$ 
6. evaluate  $P(t)$ 
7. store the best individual among  $P(t)$  and its fitness
8. while (not termination condition) do
9. begin
10.  $t \leftarrow t + 1$ 
11. create  $P(t)$  by observing the states of  $Q(t)$ 
12. evaluate  $P(t)$ 
13. update  $Q(t)$  using suitable quantum gates  $U(t)$ 
14. save the best individual among  $P(t)$  and its fitness
15. end
16. end

```

Algorithm 1 Procedure of QEA.

The method of QEA is shown in Fig. (7).

Here, $Q(t) = \{q_1^t, q_2^t, \dots, q_N^t\}$ is a population of N qubit chromosomes at generation t , and q_i^t is the i^{th} ($i = 1, 2, \dots, N$) the individual defined as [63]: $P(t) = \{X_1^t, X_2^t, \dots, X_N^t\}$ can be formulated as a set of binary solutions of the observation states of $Q(t)$, where X_i^t is the binary solution by observing q_i^t ($i = 1, 2, \dots, N$).

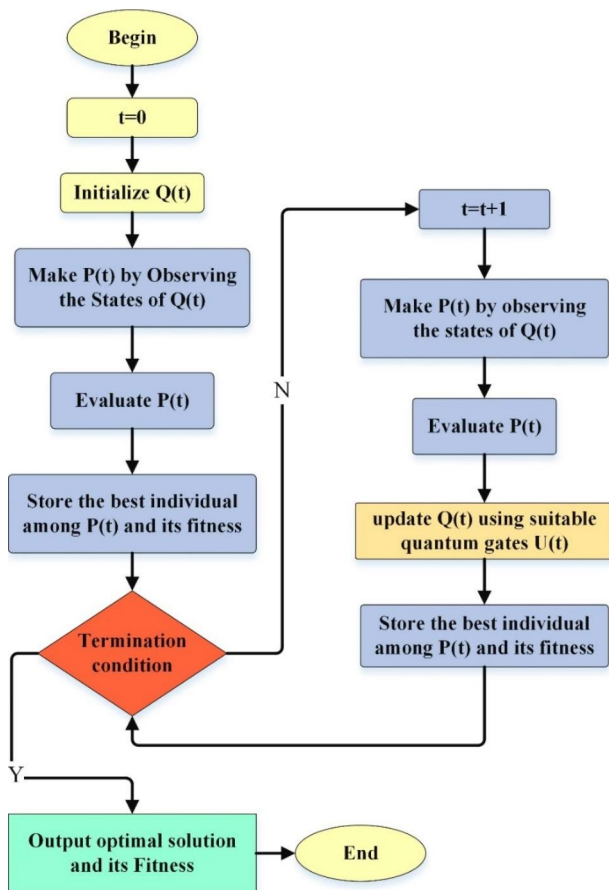
$$\begin{bmatrix} \alpha_{i1}^t & | & \alpha_{i2}^t & | & \dots & | & \alpha_{im}^t \\ \beta_{i1}^t & | & \beta_{i2}^t & | & \dots & | & \beta_{im}^t \end{bmatrix} \quad (8)$$

Each step of this algorithm has been demonstrated below [63]:

In the 'initialize $Q(t)$ ' phase, each pair of the qubit probability amplitudes, α_{ir}^t and β_{ir}^t , $r = 1, 2, \dots, m$, are initialized with $\frac{1}{\sqrt{2}}$, $\forall q_i^t \in Q(t)$. The next phase generates a set of binary solutions $P(t) = \{X_1^t, X_2^t, \dots, X_N^t\}$, where each bit of X_i^t , $i = 1, 2, \dots, N$, is formed by determining the explicit state of each qubit of q_i^t , $|0\rangle$ state or $|1\rangle$ state, according to either $|\alpha_{ir}^t|^2$ or $|\beta_{ir}^t|^2$ of q_i^t , $r = 1, 2, \dots, m$. For instance, to form an explicit state of the r th bit of X_i^t ($r = 1, 2, \dots, m$), a number h between 0 and 1 is generated randomly. Then, if $h < |\beta_{ir}^t|^2$, the r th bit of X_i^t is set to be 1; otherwise, it is set to be 0. Each solution $X_i^t \in P(t)$, $i = 1, 2, \dots, N$ is a binary string of length m , and is measured to give some measure of its fitness. The initial best solution X_{best}^t is then selected and stored among the binary solutions $P(t)$. In the while loop, the quantum gate $U(\theta)$ is applied to update $Q(t-1)$ so that the fitter states of the qubit chromosomes can be generated. The j^{th} qubit value $(\alpha_j^t, \beta_j^t)^T$ of q_i^t is updated as [63]:

$$\begin{bmatrix} \alpha_j^{t+1} \\ \beta_j^{t+1} \end{bmatrix} = U(\theta_j^t) \begin{bmatrix} \alpha_j^t \\ \beta_j^t \end{bmatrix} = \begin{bmatrix} \cos(\theta_j^t) & -\sin(\theta_j^t) \\ \sin(\theta_j^t) & \cos(\theta_j^t) \end{bmatrix} \begin{bmatrix} \alpha_j^t \\ \beta_j^t \end{bmatrix} \quad (9)$$

Fig. 7 Flowchart of QEA [63]



Here, θ_j^t is the rotation angle of q_i^t . The value of θ affects the speed of convergence. The solution may be diverged or prematurely converged to a local optimum despite being too big. The sign of θ determines the direction of the interest to a global optimum. The best solution among $P(t)$ is then selected in the next step, and if the obtained solution is fitter than the best-stored solution X_{best}^t , the best-stored solution X_{best}^t will be replaced by this current solution. Table (2) summarises the quantum-inspired EA based on various factors.

3.1.2 Ant colony optimization (ACO)

The three critical differences between QACO and ACO are that (a) the pheromone is released at the point where the ant is currently residing rather than along the entire route the ant has travelled, (b) the current ant position is represented by a group of probability amplitudes of qubits, and a group of quantum rotation gates carries out the ant move, and (c) the ant position is subjected to a quantum mutation. The ACO forms the positive feedback by exchanging pheromone. Pheromone trails leading to the overall optimum solution in the optimization space can be seen as the process runs. The experimental findings demonstrated that the QACO maintains ACO optimization capabilities and robust performance

Table 2 Summary of the Quantum-inspired EA based on various factors

Refs	Models	Objective	Role	Description	Problem
[64]	QEA	convergence speed	Increase detection accuracy	QEA has been employed to power the optimization control strategy.	the logical structure of Proportional Integral Derivative (PID)
[63]	QEA	without premature convergence and a good balance between exploration and exploitation	global search algorithm	A Q-gate is introduced as a variation operator to drive the individuals toward better solutions	COPs
[65]	QEA	convergence speed	Generate Solutions	A Quantum rotation gate (QRG) is the most generally used operator for the operation of quantum gate update, which significantly influences the efficiency of QEA.	complex function optimization problems
[66]	Quantum inspired EA (QIEA)	increase exploration	reduction Execution time	optimization technique of controller parameters	Proportional-Integral-Derivative (PID) controller
[67]	QIEA	increase exploration	Generate the initial population of n chromosomes (quantum chromosome)	Initially, the genes of all individuals in the quantum population were initialized with linear superposition.	watermark image
[68]	Quantum-Inspired Order-Based EA (QIEA-O)	convergence speed	Quantum Population Initialization	The quantum individual's representation is based on a vector of m Q-bits, producing a generic chromosome representation that can be used for a vast space of ordering problems.	Ordering optimization problem
[69]	QIEA	accelerate convergence	Quantum Population Initialization	Each element or gene of the quantum chromosome is a superposition of k q-bits	Community Detection in Social Networks
[70]	QIEA	convergence speed	Local Improvement	A new mechanism for preventing premature convergence based on rotation gate operators	High Dimension Problems
[71]	quantum-inspired competitive coevolution algorithm (QCCEA)	good balance between exploitation and exploration	near-optimal solutions in a lower number of generations	In QCCEA, the candidate solution is represented by combining a set of solution points jointly described through normal distributions.	optimization problems and multi-objective optimization problems (MOOP)

Table 1 (continued)

Refs	Models	Objective	Role	Description	Problem
[72]	Real-coded Quantum-inspired EA (RQEA)	profitable exploitation of the quantum-inspired generated solution's neighbourhood	local improvement	To improve the exploitation property of the proposed RQEA, a local search technique has been incorporated to get the best solution efficiently around a QEA's generated key at a given depth.	continuous optimization
[73]	QEA	capabilities of exploration and exploitation	reduction Execution time	Update each Q-bit individual in $Q(t-1)$ to form $Q(t)$ with the rotation gate	loading pattern design
[74]	QIEA	improved exploration and exploitation capabilities	Local Update	QIEA successfully balances the conflicting objectives of providing near-optimal solutions while keeping the computation cost low	Solving the 0–1 Quadratic Knapsack Problem
[75]	Hamming distance (HD-QEA)	convergence speed	the global optimization capability	In the HD-QEA, the diversity among individuals was considered, and an appropriate rotation angle step (RAS) was assigned to each individual according to the Hamming distance.	network coding resources optimization
[76]	QEA	convergence speed	the quantum mechanism reduces the time complexity	each chromosome in the population undergoes a quantum interference to create another population matrix for optimization	grey-level image thresholding
[77]	QCCEA	search speed	QCCEA identifies near-optimal solutions in a lower number of generations	the quantum-inspired mechanism has also been employed to enhance QCCEA's capability of creating more unique solutions to encourage vigorous competition for better solutions and, therefore, the effectiveness of the evolutionary process	numerical optimization functions
[78]	quantum-behaved EA (QBEA)	accelerate optimization	Local Improvement	To avoid premature convergence, the mutation of individuals is achieved by the QC	function optimization problems
[79]	QEA	convergence speed	Generate Solutions	Each subpopulation of $Q(t)$ independently evolves into $P(t)$, and each subpopulation of $P(t)$ is evaluated to form the best-so-far chromosomes $B(t)$, respectively.	optimization
[80]	Region-based Quantum EA (RQEA)	a right balance between exploration and exploitation	global search algorithm	the search progresses from one generation to the next; the quantum bits evolve gradually, increasing the probability of selecting regions that yield good fitness values	global numerical optimization

Table 1 (continued)

Refs	Models	Objective	Role	Description	Problem
[81]	differential real-coded quantum-inspired EA (DRQEA)	convergence speed	find optimal global solutions	two adaptive differential mutation and crossover operations are proposed and significantly enhance the global search ability of DRQEA	Hydro-thermal generation scheduling
[82]	watershed (QWC)	improved exploration and exploitation capabilities	local update	Update Q(t) by using Quantum mutation and Crossover	image segmentation
[83]	hybrid quantum clone EA (HQCEA)	convergence speed	diversity of the solution space of functions and avoid trapping into local peak	HQCEA uses the appropriate Q-gate, which operation the Q-bit should be updated as the mutation step.	Networked Control System
[84]	clonal real-coded quantum-inspired EA (CRQEA)	balancing global exploration and local exploitation	proposed CRQEA can obtain better solutions in a shorter computation time	Cloning strategy and Cauchy mutation are integrated with CRQEA to increase the diversity of the individual group of CRQEA, enhancing the global search ability of CRQEA.	short-term hydro-thermal generation scheduling
[85]	QIEA	convergence speed	the global optimization capability	Update mutation based on rotation gate	Optimization Numerical Problems
[86]	QEA	increase exploration	find optimal global solutions	Q-bits form a chromosome in the QEA, and unlike GA, which uses, for example, the operators of mutation and crossover, the population evolves based upon a variation operator known as Q-gate	continuous high-dimensional search spaces
[87]	improved quantum-inspired EA (IQEA)	The trade-off between the exploration and the exploitation	finding the optimal global solution	position i of Q-bit individual at generation t is sorted, and the ith Q-bit value is updated using the NOT gate	economic dispatch problem
[88]	Quantum-inspired Competitive EA (QUCEA)	The trade-off between the exploration and the exploitation	finding the optimal global solution	(I) the genes of an individual chromosome are represented as qubits to have a probabilistic representation of the solutions, (II) the unique utilized evolutionary operator is based on competitive learning, to be application-independent, and (III) the operator has only one setting parameter, to be tuned easily.	NP-hard problem
[89]	multi-granularity evolution quantum-inspired EA (MEQEA)	a good balance between exploration and exploitation	local improvement	MEQEA allows different chromosomes, at each generation, to have different rotation angle step values for update	resource optimization

Table 1 (continued)

Refs	Models	Objective	Role	Description	Problem
[90]	hybrid quantum-inspired EA with DE (HQEDE)	convergence speed	local improvement	HQEDE is proposed and applied to estimate the parameters of the Lorenz system	Parameter estimation
[91]	QEA	overcome premature convergence	generate solutions	individual after each updating can help the population jump out of premature convergence	optimization
[92]	adaptive-evolution-based quantum EA (AEQEA)	excellent capability in global searching	global search algorithm	AEQEA introduces an adaptive evolution mechanism (AEM), which accepts each chromosome in a population to update to a fitter location according to its situation	QoS multi-cast routing problem
[93]	Bloch Quantum-inspired EA (BQEA)	accelerate the convergence process	global search algorithm	For the rotation and mutation of qubits, two new operators are constructed based on Bloch coordinates of qubits	continuous space optimization

while maintaining quantum computation's remarkable efficiency and speed [94]. The principle pheromone level updating in QACO is to add the fitness function value of the current ant position to the pheromone, resulting in a better position holding the higher pheromone level. The principle behind updating heuristic information is adding fitness function gradient information to the heuristic, causing the position with the stronger gradient to hold the higher heuristic information level. The optimal position should have the highest fitness function value and the highest fitness function gradient value. Such a method can make each ant move in the plainness of the search process to speed up convergence and jog in the escarpment of the search process to prevent missing the optimal global solution. The QACO algorithm looks for a multi-directional path while concurrently updating the diffusion pheromone [95]. Using the ACO algorithm, the QACO calculates the number of future pheromone values based on the pheromone changes that occur along each route in the network during the initial stages. Therefore, a practical route may be created by forecasting the future pheromone values within a few repetitions. Table (3) reviews the quantum-inspired ACO based on various factors.

3.1.3 Artificial fish swarm algorithm (afsa)

In [107], a new model based on quantum PSO and AFSA (QAFSP) proposed function optimization. The proposed QAFSP uses a dynamic parameter adjustment conducive to leaping out of local optima and moving to global intensity. Experimental effects displayed that the QAFSP results in fast and dependable convergence.

To enhance the global search capability and the convergence speed of the AFSA, a unique Quantum AFSA (QAFSA) based totally on the concepts and principles of quality control, together with the quantum bit and quantum gate, has been proposed in [108]. The artificial fish's location is encoded using the angle in $[0, 2\pi]$ based totally on the qubit's polar coordinate illustration in the 2-dimension Hilbert space. The quantum rotation gate is applied to replace the artificial fish to enable the artificial fish to transport; also, the quantum non-

Table 3 Summary of the Quantum-inspired ACO based on various factors

Refs	Models	Objective	Role	Description	Problem
[95]	quantum mechanics-based convolution-based multi-directional and parallel ant colony optimization algorithm (QM-CMPACO).	convergence speed	good performance	Shortest path generation	global optimization
[96]	Quantum-stimulated of the hybrid ACO (CQACA)	global search algorithm	Fast convergence rate	CQACA is helpful to solve the slow convergence velocity and falling into neighbourhood extremum trouble	Image segmentation
[97]	QC-ACO	convergence speed	an excellent solution with a quick speed	a new approach based on QC-ACO to route navigation	global optimization
[98]	QACA	Increase Exploration	reducing computational complexity	QACA is used in sign detection of MIMO	sign detection
[99]	QACO	accelerate optimization	High performance	Discover image threshold value	Image processing
[100]	QACO	The trade-off between the exploration and the exploitation	good performance	For the low efficiency and poor performance of the ant colony algorithm in solving TSP problems	TSP problem
[101]	Ant Colony Optimization with Quantum Action (ACO-QA)	improved exploration and exploitation capabilities	find optimal global solutions	This method overcomes the shortcoming of ACO, which is easy to fall into local optimums and has a slow convergence rate in continuous space.	defection recognition
[102]	QACO	overcome premature convergence	Local Improvement	To solve collision detection in the virtual environment, this paper advanced an improved QACO algorithm.	Collision Detection
[103]	Continuous quantum ACO (CQACO)	accelerate optimization	find optimal global solutions	improve performance	global optimization
[104]	continuous quantum ACO(CQACO)	convergence speed	faster convergence rate	population diversity based on QC	global optimization
[105]	QACO	accelerate optimization	better diversity of population and ability to search and optimization	Each ant is a quantum individual	task allocation
[106]	Traveling Salesman Problem (TSP)	Increase Exploration	faster convergence rate	This mechanism offers the ability to escape from local optima	TSP optimization

gate is used to recognize the AF's mutation to speed up the convergence. Fast convergence

Table 4 Summary of the papers of proposed QGA

Refs	Algorithms	Title	Objective	Problem	Publication
[138]	Quantum-Inspired Modified GA (QIMFGA)-Based Fuzzy C-Means	Color MRI Image Segmentation Using Quantum-Inspired Modified GA-Based FCM	convergence speed	Image segmentation	Springer
[139]	Quantum-Inspired GA (QIGA)	Use of A Quantum GA for Coalition Formation in Large-scale UAV Networks	convergence speed	Task allocation	Sciencedirect
[140]	Quantum GA with Rotation Angle Refinement (QGARAR)	Quantum GA with Rotation Angle Refinement for Dependent Task Scheduling on Distributed Systems	depreciation of the makespan value	task scheduling	IEEE
[141]	Quantum-inspired GA-based Load Balancing Strategy (QLBS)	Quantum GA-based Scheduler for Batch of Precedence Constrained Jobs on Heterogeneous Computing Systems	effective exploitation	Batch allocation	Sciencedirect
[142]	quantum GA (QGA)	Path Planning for a Space-Based Manipulator System Based on Quantum GA	convergence speed	optimization problem	Hindawi
[143]	Hybrid Quantum-Inspired GA (HQIGA)	An improved Hybrid Quantum-Inspired GA (HQIGA) for real-time scheduling tasks in a multiprocessor system	effective exploitation	Real-time task scheduling	Sciencedirect
[144]	QGA	Based on Two Element Neighborhood Search Quantum GA to Solve the Vehicle Scheduling Problem	improving global search	Vehicle Scheduling Problem	IEEE
[145]	QGA	A Novel Improved Quantum GA for Robot Coalition Problem	faster convergence rate and more substantial global search capability	Robot Coalition Problem	IEEE
[146]	quantum algorithm (QA) based on a discrete-time quantum walk (QWGA)	Quantum walk-based GA for 0–1 Quadratic Knapsack Problem	rapid convergence and avoidance of local optima	0–1 Quadratic Knapsack Problem	IEEE
[147]	improved quantum GA (IQGA)	Improved Quantum GA in Application of Scheduling Engineering Personnel	global optimal solution	Scheduling Engineering Personnel	Hindawi
[148]	QGA	A New Approach of Quantum-Inspired GA for Self-Generation of Fuzzy Logic Controller	enforce the convergence	Fuzzy Logic Controller	IEEE
[149]	QGA	The Improvement of Quantum GA and Its Application on Function Optimization	effective exploitation	Function Optimization	Hindawi
[150]	IQGA	To Solve the Job Shop Scheduling Problem with the Improve Quantum GA	accelerate the convergence rate	Job Shop Scheduling	IEEE

Table 3 (continued)

Refs	Algorithms	Title	Objective	Problem	Publication
[151]	QGA	Quantum GA for Mobile Robot Path Planning	capabilities of searching and convergence	Robot Path Planning	IEEE
[152]	quantum-inspired GA for k-means clustering (KMQGA)	A quantum-inspired GA for k-means clustering	effective exploitation	Clustering	Sciencedirect
[153]	competitive co-evolutionary QGA (CCQGA)	A novel competitive co-evolutionary quantum GA for stochastic job shop scheduling problem	avoid premature convergence	stochastic job shop scheduling problem	Sciencedirect
[154]	general quantum GA (GQ-GA)	Reactive power and voltage control based on general quantum GAs	global optimal solution	Reactive power control	Sciencedirect
[155]	IQGA	An Improved Quantum GA	avoid premature convergence	Function Optimization	IEEE
[156]	Novel Parallel Quantum GA (NPQGA)	A novel parallel quantum GA for stochastic job shop scheduling	rapid convergence and good global search capability	stochastic job shop scheduling	Sciencedirect
[157]	HQGA	Quantum-inspired GAs for flow shop scheduling	A proper balance between exploration and exploitation	flow shop scheduling	Springer
[158]	QGA	Quantum Model of GA	capabilities of searching and convergence	Optimization	IEEE
[159]	real-coded QGA (RQGA)	Hybrid GA based on QC for numerical optimization and parameter estimation	faster convergence rate and more robust global search capability	numerical optimization problems	Sciencedirect
[160]	QGA	Cultural Operators for a Quantum-Inspired Evolutionary Algorithm Applied to Numerical Optimization Problems	avoid premature convergence to local minima	numerical optimization problems	Springer
[161]	QGA	A Novel Immune Quantum-Inspired GA	balance ability between global search and local search	COP	Springer
[162]	parallel quantum-inspired GA (PQGA)	Parallel Quantum-inspired GA for COP	rapid convergence and good global search capability	COP	IEEE
[163]	QGA	Genetic QA and its Application to COP	Rapid convergence and good global search capability	COP	IEEE
[137]	QGA	Quantum-inspired GA	balance ability between global search and local search	optimization	IEEE

Table 5 Summary of the papers related to the proposed QPSO

Refs	Models	Title	Objective	Problem	publication
[175]	QPSO	an improved quantum-behaved PSO (QPSO) method to estimate the model parameters	Increase detection accuracy	Optimization	Sciencedirect
[176]	QPSO	A new QPSO for controller placement problems in software-defined networking	Convergence rate	Controller placement problem	Sciencedirect
[177]	QPSO	Diversity-preserving QPSO for the multidimensional knapsack problem	global optimal solution	Multi-dimensional	Sciencedirect
[178]	QPSO	QPSO with guided exploration for function optimization	Exploration and exploitation	Engineering design	Sciencedirect
[179]	QPSO	Fracture performance using QPSO	better convergence and robustness	prediction	Sciencedirect
[174]	quantum Delta PSO (QDPSO)	PSO with Particles Having Quantum Behavior	convergence speed	benchmark functions	IEEE
[180]	quantum PSO clustering algorithm to improve network lifetime (QPCINL)	Quantum PSO Algorithm for Clustering in WSN to Improve Network Lifetime	convergence speed	Clustering	Springer
[181]	QPSO	The Utilization of Quantum Inspired Computational Intelligent in Power Systems Optimization	a proper balance between the exploration and the exploitation	power systems optimization	Springer
[182]	QPSO	Optimal Distributed Generation Allocation Using Quantum Inspired PSO	convergence speed	power system problem	Springer
[183]	QPSO and fuzzy clustering (QPSOFC)	A sonar image segmentation algorithm based on quantum-inspired PSO and fuzzy clustering	global optimal solution	image segmentation	Springer

Table 4 (continued)

Refs	Models	Title	Objective	Problem	publication
[184]	QPSO	A Quantum Particle Swarm-Inspired Algorithm for Dynamic Vehicle Routing Problem	less parameter involved in the iteration, faster computing speed and more vital ability of global searching	Vehicle Routing Problem	Springer
[185]	QPSO	A Model for Resource Constraint Project Scheduling Problem Using Quantum Inspired PSO	convergence speed	The resource-constrained project scheduling problem	Springer
[186]	quantum computation concepts and rough set theory (QCSIA-FS)	A cooperative swarm intelligence algorithm based on quantum-inspired and rough sets for feature selection	a good trade-off between the exploration and the exploitation	feature selection	Sciencedirect
[187]	map-reduce two phases quantum behaved fuzzy rule PSO (MR-TP-QFPSO)	MR-TP-QFPSO: map-reduce two phases quantum fuzzy PSO for feature selection	a good trade-off between the exploration and the exploitation	feature selection	Springer
[188]	QPSO	A hybrid quantum-induced swarm intelligence clustering for the urban trip recommendation in an intelligent city	-	Recommender system	Sciencedirect
[189]	Reinforced Multi-Objective quantum-behaves PSO (RMO-QPSO)	Multi-objective design of state feedback controllers using reinforced quantum-behaved PSO	global convergence	Linear quadratic regulator (LQR)	Sciencedirect
[190]	improved QPSO (IQPSO)	Kinematics Parameters Identification for IRB 1400 Using Improved Quantum Behaved PSO	convergence speed	Robot positioning accuracy	Springer
[191]	QPSO Technique for Multi-level Colour Image Thresholding (QIPSOMLTIC)	Efficient quantum-inspired metaheuristics for multi-level true-colour image thresholding	convergence speed	Colour image thresholding	Sciencedirect

Table 4 (continued)

Refs	Models	Title	Objective	Problem	publication
[192]	cellular QPSO (CQPSO)	A decentralized quantum-inspired PSO algorithm with a structured cellular population	balancing the global search and local search to enhance the search efficiency	different neighbourhood shapes and grid shapes	Sciencedirect
[193]	QPSO	Calibrating the RZWQM2 model using quantum-behaved PSO algorithm	initialization	automatic calibration	Sciencedirect
[194]	krill herd QPSO (KH-QPSO)	A hybrid method based on krill herd and quantum-behaved PSO	global optimum	solving continuous optimization problems	Springer
[195]	Quantum Inspired PSO (QIPSO)	Multi-level thresholding using quantum-inspired metaheuristics	convergence speed	Image thresholding	Sciencedirect
[196]	QPSO	A quantum-behaved PSO with memetic algorithm and memory for continuous non-linear large-scale problems	global searchability	The constant non-linear large-scale problem	Sciencedirect
[197]	QPSO	Constrained Multi-objective Optimization Using a Quantum Behaved Particle Swarm	A balance between searching in infeasible and feasible regions	constrained multi-objective problems	Springer
[198]	QPSO sequential quadratic programming (QPSO-SQP)	A Hybrid Quantum-Inspired Particle Swarm Evolution Algorithm and SQP Method for Large-Scale Economic Dispatch Problems	convergence speed	economic dispatch problems	Springer
[199]	QPSO	A Multi-objective Resource Assignment Problem in Product Driven Supply Chain Using Quantum Inspired Particle Swarm Algorithm	A good balance between exploitation and exploration	multi-objective optimization problem	Springer

Table 4 (continued)

Refs	Models	Title	Objective	Problem	publication
[200]	QPSO	T-S Fuzzy Control for Magnetic Levitation Systems Using Quantum PSO	global optimum	Optimization	IEEE
[201]	QPSO	Quantum-behaved PSO algorithm for economic load dispatch of power system	increasing population diversity	economic load dispatch	Sciencedirect

and accuracy worldwide seek the capacity to signify the overall performance of QAFSA. The experimental outcomes show that the performance of QAFSA is significantly enhanced compared to that of the standard AFSA.

3.1.4 Bat algorithm (BA)

Dey et al. [109] added a quantum-stimulated framework to the BA to automatically cluster picture datasets. These paintings' purpose is to determine the optimal number of clusters from a photo dataset on the run. A contract has been made between the quantum-inspired BA (QIBAT) and its classical counterpart. The proposed QIBAT has been found to excel in its classical counterpart as it can distinguish the ideal number of clusters mechanically from a picture dataset with a shorter computation time. Computational experiments have been carried out on four Berkeley image datasets.

In [110], a novel quantum-behaved BA with the mean best position (QMBA) route has been offered. In QMBA, every bat's placement is specially updated with the help of the solution obtained within the early stages of a search. The late search also depends on the mean quality position, which can increase the convergence speed of the algorithm. This approach results in fast convergence, simplicity, and easy application of original BA and will increase the variety of population and improve solutions' accuracy. Twenty-four test functions were evaluated and compared with other versions of BAs for numerical optimization. The results indicate that QMBA is efficient and straightforward and can provide more accurate solutions.

3.1.5 Bee colony optimization (BCO)

Bloch spherical coordinates based on quantum artificial bee colony optimization (BQBCO) [111] to image segmentation have been proposed. Image segmentation is a critical step in many computer vision applications, and threshold-based methods are extensively used for this purpose. The central issue in threshold segmentation is how to get an acceptable threshold. The two-dimensional linear cross-entropy of the image is used as a fitness feature. Meanwhile, the picture threshold segmentation problem has been researched using BQBCO. The experimental results showed that the BQBCO is excellent in image threshold and satisfying and fast image segmentation.

The fuzzy c-means (FCM) algorithm is a general approach [112–115]. The FCM set of rules and its research are usually tormented by choice of initial values and noise facts and might easily fall into excellent neighbourhood value. To overcome these drawbacks of FCM, [116] proposed an algorithm primarily based on multi-chain quantum BCO (MQBC-FCM). In this algorithm, more than one chain encoding approach is introduced to the BCO to recommend the MQBC set of rules. Then, MQBC is used to look for the premier preliminary clustering centres. The proposed MQBC-FCM has been used on artificial datasets and image segmentation, and its function is contrasted with numerous algorithms. The experimental results have indicated that the offered MQBC-FCM has accurately improved the position of the clustering algorithm.

To accurately solve COPs, a membrane-stimulated quantum BCO (MQBCO) [117] was proposed for scientific computing and engineering applications. The offered MQBCO algorithm employs a robust discrete optimization algorithm's membrane computing theory on quantum BCO (QBCO). The global convergence function of MQBCO is proved using Markov's theory, and the validity of MQBCO is established by checking out the classical benchmark operations. Then, the recommended MQBCO algorithm is used to solve decision engine problems of the cognitive radio system. Simulation experiment; 2nts below unique verbal exchange scenarios illustrate that the balance between three-goal capabilities and the adapted parameter configuration is steady with the weights of three normalized goal operations.

3.1.6 Bacterial foraging optimization algorithm (bfoa)

In [118], a novel swarm intelligence optimization technique that integrates BFO with QC has been offered with Quantum BFO (QBFO). Numerous assessments were based on benchmark features, including a multi-peak operation to evaluate the optimization performance of the proposed QBFO. Numerical results indicate that the proposed QBFO has more valuable features in terms of convergence rate, balance, and the capacity to find the most suitable solution than the original BFO and quantum GA. Moreover, they assessed the employment of their proposed QBFO for cognitive radio spectrum allocation. The results of solving the spectrum allocation problem show that QBFO is better than colour-sensitive graph colouring (CSGC) and QGA. In [119], a novel swarm intelligence optimization technique that integrates BFO with QC was proposed using the quantum BFO (QBFO) algorithm. A probabilistic multi-qubit that could constitute a linear superposition of states in the search area is used to represent a bacterium so that the quantum bacteria illustration has a higher function of population diversity. A quantum rotation gate is adjusted to simulate the chemotactic step to drive the bacteria towards better answers. They implemented the proposed QBFO to solve the TSP. The results suggest that the proposed QBFO shows better convergence behaviour without premature convergence and has more powerful properties in the convergence rate, stability, and search for the ideal global answer than ACO and QGA.

To eliminate the delay in optimization and to further increase the performance of BFA, a bacterial foraging algorithm based totally on QPSO (QPSO-BFA) was offered [120]. The new algorithm was proposed to combine the benefits of both algorithms to get higher optimization values. The simulation outcomes on eight benchmark capabilities indicated that the proposed QPSO-BFA is superior to the BFA, QPSO and BF-PSO.

A quantum-stimulated bacterial swarming optimization (QBSO) algorithm based on the BFO problem was proposed to solve discrete optimization [121]. The proposed QBSO employs the QC on BFO and benefits both quality control theory and BFO. Also, they exploited the swarming pattern of birds in blocks added in PSO. Then, they evaluated the function of the offered QBSO via four classical benchmark features. The obtained outcomes showed that the designed QBSO is superior to a few preceding intelligence algorithms in convergence rate and accuracy.

3.1.7 Cuckoo search algorithm (CSA)

Minimizing the make-span in the no-wait flow shop scheduling problem (NWFSP) is broadly employed in numerous industries. However, it is an NP-hard problem. A novel quantum-stimulated CS (QCCS) was proposed to remedy this problem [122]. The QCCS includes these three phases: (I) A Quantum illustration state solution, (II) A Quantum-Stimulated Cuckoo Search Differential Evolution (QCS-DE) search, and (III) A Local Neighborhood Search (LNS) algorithm. The QCCS algorithm was assessed using Rec and car benchmark samples and compared with the state-of-the-art algorithms, including GA-VNS, HGA, and TS-PSO. Furthermore, the in-depth statistical analysis demonstrated the effectiveness of the modern proposed QCCS. The numerical results affirmed that the proposed QCCS has strong optimization capability and can effectively solve the NWFSP on a small and medium scale.

In [123], worldwide best steered Quantum stimulated CSA (GQICSA), has been proposed for acquiring the optimized set state coefficients to enforce Finite Impulse Response (FIR) clear out. Adder cost state a filter is estimated after quantizing the filter coefficients accompanied via standard Sub-expression removal (CSE). They discovered from the simulation outcomes that reducing word duration state coefficients does not make the filters fail to obtain the perfect frequency reaction. Furthermore, filters advanced using GQICSA outperform the benchmark filters designed by the Parks McClellan algorithm in state stop-band attenuation. The results indicated that GQICSA improves over various conventional algorithms, including CSA, and surpasses changed model state CSA. Quantum-inspired CSA (QICSA) updated the use of quantum principles for optimizing filter-out coefficients to design lower hardware costing filters compromising the filter responses and efficiency. GQICSA also presents significant upgrades compared to CSA and QICSA in today's stop-band attenuation and execution time for higher-order filter design. The Performance state GQICSA over QICSA and traditional CSA has been proved with 16 benchmark operations.

The optimization of web services composition is one of the most exciting and demanding situations at present. Because of that, the composition of web services has become a promising technology in the diffusion of regions. Furthermore, selecting the best web services composition will become an NP-hard optimization problem that may be modelled as a multi-dimensional knapsack problem. In [124], the abased Quantum method stimulated CSA to discover the quality web services Composition (QICSA-WSC) to improve the web. More precisely, they proposed two items: in the first one, we outline the perfect quantum illustration for solutions, while in the second one, they proposed a new randomized heuristic that produces answers starting with the consumer's request. To demonstrate QICSA-WSC, they applied it as a model prototype and carried it out on a text translation case study. The obtained results are very encouraging and express the feasibility and effectiveness of the proposed method.

3.1.8 Differential evolution (DE)

In [125], in line with the traits of the shipboard energy system, a complete fault reconfiguration was constituted, and a couple of objective capabilities were included by the analytic hierarchy method (AHP). A kind of quantum DE algorithm, TQDE, was proposed for solving the version, and a triple code for quantum computation was designed to satisfy the particular demand of the shipboard strength system for coding. In the TQDE algorithm, the Q-bit state is updated via DE, so the difficulty in the calculation by table lookup is averted. Because of the quantum superposition state, the population variance is maintained, and optimized efficiency is promoted. Furthermore, individual migration is found using the dynamic rotation gate method to prevent premature convergence. A simulated analysis was conducted for the real-world example to prove the technique's effectiveness and lay a foundation to apply it in the real world.

Yin et al. [126] designed a new implementation for FS's concept model. To solve this problem, they proposed a new algorithm called the free search with Adaptive DE Exploitation and Quantum-stimulated Exploration (ADEQFS). ADEQFS aimed to design a new mutation method using adaptive DE strategies, concepts, and real-coded quantum-stimulated EA principles. Furthermore, they used the conventional DE scheme's crossover operation to alleviate the proposed algorithm's untimely convergence. The convergence evaluation of the proposed ADEQFS was also presented in this paper. They provided proof of convergence by exploiting the Markov chain model. Thirty-four optimization test functions with distinctive mathematical traits were employed as benchmark sets to check the overall performance of ADEQFS. The numerical results showed an increase in convergence rate and computation reliability.

Zheng and Yamashiro [127] proposed a unique quantum differential evolutionary algorithm (QDEA) primarily based on the fundamental QEA for the permutation flow shop scheduling problem (PFSP). Here, the quantum chromosomes were encoded and decoded by applying the quantum rotating angle and a straightforward strategy named the most significant rotating angle cost rule to determine task series based on jobs. First of all, quantum information was applied for the illustration of PFSP. They combined the DE method's benefits, the variable neighbourhood search, and QEA by adopting the DE to update the variable neighbourhood search and quantum gate to elevate the local search function. They adopted QDEA to decrease the make-span, general flow time, and the most recent jobs and make the simulations. The results and comparisons with other algorithms primarily based on well-known benchmarks illustrated the effectiveness of the proposed QDEA.

In [128], a new variable ordering method referred to as QDEBDD was proposed to lessen the Binary Decision Diagram (BDD) size. The scale of BDD depends to no small extent on the order of operation variables. Unfortunately, the search for ordering the exceptional variables has been confirmed as NP-hard. In this study, the variable ordering problem was definite as an optimization problem for which a new framework relying on QC was proposed. Their contribution is defining the proper quantum illustration scheme that successfully applies a few quality control standards to the BDD problem. This representation scheme was embedded within a DE algorithm, leading to an efficient hybrid framework that improved stability between the search procedure's exploration and exploitation competencies.

3.1.9 Firefly algorithm (FA)

In [129], a multi-population extension of FA was proposed to solve multi-objective dynamic optimization problems. The proposed FA changed the usage of fitness operation evaluations more efficiently. The quantum particles that implicitly carry out a local search within quantum clouds were employed to preserve variety and increase solution quality. The proposed algorithms were compared to the well-known methods in the related literature. This experimental study and its statistical tests affirmed that the roulette-based choice method operates better than sequence-based selection, specifically in instances with extra peaks. Moreover, it is worth stressing that even though the effects are not the best-reported results, the proposed methods appear to be favourable compared to the well-known strategies. As illustrated in the present contribution, the prioritizing methodology of sub-swarms is an essential component that might affect the applied multi-population-based algorithm's overall performance.

In [130], FA was improved by quantum technique integration and was exploited as an optimization means for bridge flutter analysis. The proposed algorithm was formed on the integration into the FA of the quantum GA (QGA). Using the Quantum Genetic FA (QGFA), the results were reviewed with FA and QGA for validation. (1) The population of the FA was quantized through the dual encoding format. They defined an update method for the quantum rotation gate, provided a type of the hybrid FA, and searched the optimization model's optimization with the combining FA. The quantum rotation gate update method was defined to update the population situation, and the quantum crossover and mutation operation was presented to improve the FA's global searchability. (2) The QGFA, FA, and QGA were used to analyze three cases. The findings indicated that the QGFA leads to higher optimization results and stability. The CPU time of the QGFA was more than FA and QGA. (3) The entire computing process of the QGFA was free of manual intervention. The algorithm had good practicality, numerical balance, and high precision in the critical state search. In [131], a strategy was recommended to optimize the Power Quality (PQ) and the reliability of distribution systems by applying optimal Network Reconfiguration (NR). Optimal NR was used separately to a system in a certain period to reduce the number of propagated voltage sags (*Nsag*) and other reliability indexes such as the average system interruption frequency index, maintained average interruption frequency index, and the momentary average interruption frequency index. The quantum-stimulated dual FA (QBFA) was used to find the optimal NR. The QBFA operation for using optimal NR to reduce *Nsag* was first compared with other constituted optimization methods such as the standard binary FA and GSA. Simulation results illustrated that the proposed optimum NR effectively increases system reliability and PQ.

3.1.10 10. Fireworks algorithm (FA)

A new time slot structure was designed for energy harvesting (EH) cognitive radio (CR) to achieve high energy efficiency and maximal throughput. Assuming the CR system with EH and cooperative relay, the Best Cooperative Mechanism (BCM) was offered for CR with EH. A quantum FA (QFA) was designed to resolve the maximum throughput and EH difficulties to get the optimal estimation function. The proposed cooperative mechanism was referred to as QFA-BCM [132]. The proposed QFA combined the benefits of quantum computation theory with the FA. Therefore, the QFA was capable of acquiring the

optimal solution, and its convergence performance was proved. Simulation results proved that the QFA-BCM method is superior to the previous non-cooperative and cooperative mechanisms. Also, in [133], the opposition-based quantum FA (OQFA) was presented for solving continuous optimization problems. OQFA combined FA and two optimized operators: opposition-based learning (OBL) and QC theory. The OBL operator can speed up the convergence rate of an algorithm by retaining the better solution, and the QC theory can improve the ability to search and increase the exploration efficiency of the solution space. Since OQFA had both OBL and QC features, it showed an excellent possibility for finding a global optimum and avoiding premature convergence. Experimental results on five test functions indicated that OQFA outperforms the Cultural Algorithm (CA), PSO, and FA regarding convergence rate and accuracy.

3.1.11 11. Flower pollination algorithm (FPA)

FPA is a new metaheuristic optimization algorithm, but its population diversity and convergence precision can become limited in some applications. QC has been applied to optimization and structural engineering design problems to improve its exploitation and exploration capabilities. It is modelled in cooperative relay networks considering co-channel interference to obtain an optimal selection scheme with multiple potential relays in cooperation with relay networks and source-destination pairs. It is an integer optimization problem for choosing the proper relay nodes to acquire the optimal system function. To get the ideal estimation function with low computing complexity, Quantum FPA (QFPA) was recommended to resolve the difficulties of ideal relay selection, and the proposed optimal relay selection scheme was referred to as QFPA based on Optimal Relay Selection (QFPA-ORS) [134]. The recommended QFPA combines the benefits of the quantum evolutionary theory and FPA. Therefore, QFPA can search for the ideal solution. Simulation results indicated that the QFPA0-based relay choosing scheme could find globally optimal solutions compared to other relay selection schemes based on conventional algorithms and operate better than different intelligent algorithms.

To resolve the multi-parameter optimization problem of the infinite impulse response (IIR) digital filter design, a cultural quantum-stimulated FPA (CQFPA) was offered [135]. The proposed CQFPA is a multi-dimensional search algorithm for the optimization of real numbers. It exploits the quantum and cultural evolution mechanisms to update the quantum pollens, improving an algorithm's ability to find optimal solutions. Computer simulations indicated that the IIR digital filters formed on the CQFPA are superior to previous filters based on the quantum-behaved PSO (QPSO) and adaptive quantum-behaved PSO (AQPSO) of convergence speed and optimization results. Simulation experiments also illustrated the effectiveness and superiority of the CQFPA. In [136], a new hybrid optimization strategy called the quantum-behaved FPA (QFPA) was proposed. The method combined the standard FPA with the quantum-behaved search mechanism to improve global search capability and accuracy. The simulation experiments indicated that the recommended QFPA algorithm could optimize convergence speed and solution quality successfully.

3.1.12 12. Genetic algorithm (GA)

The Quantum-inspired GA (QGA) is based on such concepts and principles of QC as qubits and the superposition of states [137]. Compared to classical GAs, QGA has many advantages, such as the automatic ability to balance global and local search, among others. However, the process through which QGA evolves the qubit chromosomes is probabilistic, which gives the individuals an evolutionary chance and causes inevitable degeneracy. Furthermore, QGA neglects the assistant function of the characteristics of knowledge in a pending problem. Unfortunately, the function of simple QGA is often not satisfactory and is easily trapped in local optima, leading to premature convergence. The Q-bit search with quantum mechanism and genetic search with evolution mechanism must be well-coordinated, and the exploration and exploitation behaviours must be well balanced. That is because it should change according to the type of problem, parameters, and quantum bits. The summary of the papers offering QGA is shown in Table (4).

3.1.13 13. Gravitational search algorithm (GSA)

In the analysis of cloud reliability, failure prediction is of critical importance. Recently, a novel kernel learning method referred to as the relevance vector machine (RVM) has been widely used to solve nonlinear anticipating problems and verified to do well in many cases. However, it continues to be a great challenge to acquire the optimal RVM parameters. The Quantum-inspired Binary GSA (QBGSA) was proposed to solve this problem in [164]. Moreover, a hybrid model of RVM with improved QBGSA called IQBGSA-RVM [165] was offered to predict cloud services' failure times. To evaluate the effectiveness of IQBGSA-RVM in failure prediction, its predicting function was compared with that of the following algorithms, all of which employ RVM: disordered GAs, dual GSA, binary PSO, quantum-inspired binary PSO, and standard QBGSA. The experimental results indicated that the IQBGSA-RVM model is more compatible than the others.

Over the past few years, wireless sensor networks (WSN) have been exploited extensively in applications with unique features and requirements. In [166], a new version of the quantum-stimulated GSA was offered as a further optimization methodology, well-suited for quadrivalent encoding problems. The recommended strategy is adapted in a WSN in precise agriculture to find an optimal and adaptive design of WSN to optimize the energy consumption and develop the life span of the network, taking into account the communication boundaries and application-specific necessities. The offered method function, QQI-GSA, is compared with the binary GA (BGA) and binary PSO (BPSO). The experimental results indicated that the WSN designed by QQIGSA has the highest function in fulfilling the requirements and prolonging a network's lifetime.

In [167], a novel GSA using quantum mechanics theories was proposed to create a quantum-inspired GSA (QIGSA). Such approaches in the proposed QIGSA provide a powerful strategy to diversify the algorithm's population and improve its function in preventing premature convergence to local optima. The effectiveness of the QIGSA in solving various benchmark optimization functions was confirmed by the simulation results and comparison with nine state-of-the-art algorithms.

3.1.14 14. Glowworm swarm optimization (GSO)

A new intelligence algorithm called the quantum-inspired GSO (QGSO) was offered [168]. By combining the GSO, quantum evolutionary theory, and quantum coding, the quantum state and binary state of the quantum glowworms can be appropriately developed by the simulated quantum rotation gate. The conventional benchmark functions were used to examine the effectiveness of QGSO. The recommended QGSO algorithm is undoubtedly an effective optimization algorithm, which results in better convergent accuracy and speed. Thus, it is used to resolve thinned array optimization problems. Simulation results indicated that the proposed thinned array method, based on QGSO, is superior to the thinned array methods based on previous conventional intelligence algorithms. The offered thinned array method based on QGSO can search for the optimal global solution of thinned arrays.

3.1.15 15. Grey wolf optimization (GWO)

In [169], a new direction-finding method for the biostatic multiple-input multiple-output (MIMO) radar in the impulse noise was offered. The strategy can suppress the impulse noise using complete norm normalization and perform direction finding better via the weighted signal subspace fitting algorithm. A quantum-inspired GWO was applied to obtain the ideal global solution to solve this method's objective function. The recommended approach based on QGWO can resolve the direction-finding complexities of the biostatic MIMO radar. Monte-Carlo's experiments proved the robustness and superiority of the proposed QGWO for locating independent and cohesive sources with fewer snapshots in the impulse noise than some current direction-finding models in a series of frameworks. Moreover, they presented the Cramer-Rao bound (CRB) angle estimation for the biostatic MIMO radar in the impulse noise, which generalizes the Gaussian CRB for performance analysis.

3.1.16 16. Harmony search algorithm (HSA)

In [170], a new metaheuristic algorithm called Multi-scale Quantum HSA (MQHSA) was proposed based on quantum motion. The calculation accuracy of MQHSA is adjustable. This paper has detailed and well interpreted the physical model and mathematical analysis of the offered MQHSA. The structure of MQHSA is straightforward and includes exclusively two phases: the multi-scale process and the quantum harmonic oscillator process. Experiments were carried out to validate the effectiveness and efficiency of MQHSA by applying it to 30 well-defined benchmark functions. The researchers also compared MQHSA with well-known metaheuristic algorithms, such as GA, SA, PSO, and quantum PSO (QPSO). The comparative results indicated the competitive and superior performance of the proposed MQHSA in convergence speed and optimal solution accuracy. In [171], a new hybrid natural algorithm based on HSA and QC was proposed. The fundamental trait of the offered algorithm called Quantum Inspired HSA (QIHSA) is integrating the quantum representation scheme in the main HSA that authorizes the effective use of several quantum-inspired operators such as interference and measurement. This hybridization between quantum stimulated computing and HSA creates an efficient hybrid framework that better balances exploring HSA and the exploitation capabilities of the quantum-stimulated measuring. Experiments

on knapsack problems indicated the effectiveness of the proposed QIHSA and its ability to achieve good quality solutions.

3.1.17 17. Invasive weed optimization (IWO)

In [172], an advanced QEA was proposed to solve continuous optimization problems primarily based on the IWO algorithm. The experimental results confirmed that the proposed Q-IWO algorithm is an outstanding model for solving complicated optimization problems. It can also be employed for different complex real-global optimization problems. The proposed Q-IWO showed superior performance compared to the conventional IWO algorithm.

3.1.18 18. Multi-verse optimization (MVO)

A new hybrid algorithm called the quantum MVO (QMVO) was proposed in [173]. The proposed QMVO is formed based on the QC and MVO algorithms. The essential traits of the quantum strategy and MVO were employed in a new algorithm to obtain the optimal trade-off between exploration and exploitation. The QMVO algorithm depended on adopting a quantum agent of the search space and integrating the quantum interference and operators in the multiverse optimization algorithm to obtain the optimal solution of the objective function. The performance of the QMVO algorithm was evaluated by exploiting 50 unimodal and multimodal benchmark operations. The experimental results showed that the proposed algorithm is superior in solving complex numerical optimization problems. Furthermore, results indicated that the proposed QMVO is a robust optimization algorithm compared with other methods.

3.1.19 19. PSO

In [174], various conventional PSOs named Quantum-behaved PSO (QPSO) was proposed. In searching for possible solutions quickly and globally, the QPSO has better performance since its aggregation trait, which is the updating method of particle positions, is quite different from that of the PSO. In many studies where QPSO has been introduced, the benefits of QPSO in the convergence rate and global searching ability compared with PSO have been confirmed. The QPSO algorithm has successfully been applied in genetic code synthesis, economic, environmental hydrothermal energy system scheduling, image processing, financial analysis, network communication, etc. The fundamental way QPSO differs from normal PSO is that the precise values of x and v cannot be known. As a result, a process is known as “collapsing” maps the likelihood of discovering a particle at a specific location in the quantum search space to that particle’s actual position in the solution space. In Table (5), the summary of the papers related to the proposed QPSO has been presented.

The key benefit of QPSO is that the algorithm can react quickly to dynamic events and converges to the global optimum in a limited number of iterations. This is mainly attributable to the coordinated action of neutral and quantum particles, which balances the algorithm’s capacity for exploitation and exploration. To prevent the algorithm from being stuck at a local optimum, quantum particles maintain diversity. Overcrowding in a single place is eliminated as the many regions of the search area are tracked by the QPSO, and the algorithm’s capacity to adapt to changing settings is improved. The well-known 0–1 multi-

mensional knapsack problem has been solved using a diversity-preserving QPSO technique [177]. A diversity-preserving population updating technique to maintain a healthy diversity of particle swarms and a variable neighbourhood descent procedure performed probabilistically to support search intensification are two new original characteristics of the proposed algorithm.

3.1.20 20. Artificial immune algorithm (AIA)

Through embedding a capacity evolution system into the affinity function calculation of the multi-elitist immune clonal optimization and updating the cluster middle primarily based on the distance matrix, the Multi-Elitist Immune Clonal Quantum Clustering algorithm (ME-ICQC) was proposed in [202]. In the proposed ME-ICQC, the elitist population consists of individuals with an excessive affinity for playing dominant roles in the evolutionary system. This method can assist in locating the global top-quality or near-optimal solutions for maximum examined responsibilities. The variety of populations may be well maintained through the known subgroup evolution of ME-ICQC. Dissimilar mutation techniques or crossover operators carry out these exceptional features—the bi-group exchanges the records of excellence antibodies using the hypercube coevolution operation. Compared with modern algorithms, the ME-ICQC achieves improved clustering accuracy with greater reliable convergence, but it is not considerably better than different optimization strategies blended with QC. Additionally, the experimental results showed that ME-ICQC performs well on multi-class, parameter-sensitive, and large datasets.

In [203], an efficient Quantum AIA (QAIA) was proposed for the Hybrid Flow Shop Problem (HFSP). The objective was to discover an outstanding job collection that reduces the mean flow time. Because HFSP has been proven to be NP-hard in a strong sense, even in the case of two levels, the AIA and QA were used to solve the problem. To enhance the overall performance of AIA, an effective AIA with new adaptive crossover and fractional elements mutation operators was proposed. A randomly replacing method was employed to promote the population range of QA, called RRQA. To acquire better results, the paper proposed a QAIA, which combines AIA with QA to optimize the HFSP. Furthermore, all the enhancements into QIA by ARRQAIA possess the merits of global exploration, robustness, and fast convergence. The simulation results indicated that the proposed QAIA extensively enhances the overall performance of AIA.

In [204], a novel Hybrid QAIA (HQAIA) was proposed for locating the Pareto optimal solutions MOOP based totally on AIA and QC. Inside the proposed HQAIA, there were particular characteristics as follows. First, the encoding method was based on a Q-bit illustration; therefore, a chaos-based method is usually recommended to initialize the population. Second, a brand-new chaos-based rotation gate and Q-gates were offered to carry out mutation and enhance the population's quality. In the end, in particular, a brand-new truncation algorithm with similar individuals (TASI) was utilized to maintain the range of the population. Additionally, a new choice operator was proposed to create a brand-new population-based totally on TASI. The Simulation results conducted on six common problems (ZDT6, CP, SP, VNT, OSY and kit) indicated that the proposed HQIA is capable of discovering a higher percentage of solutions and has a higher convergence rate near the right Pareto-optimal front as compared to the immune vector algorithm and the elitist non-dominated sorting genetic system (NSGA-II).

3.1.21 21. Immune clonal algorithm (ICA)

In [205], Shang et al. proposed an approach to the large-scale Capacitated Arc Routing Problem (CARP) called Quantum-stimulated ICA (QICA-CARP). This algorithm combines the characteristic of ICA and quantum computation ground at the quantum superposition and the qubit. The mutation method of the quantum rotation gate speeds up the convergence of the original clone operator. Furthermore, the quantum crossover operator enhances data exchange and will increase the range of the population. Also, it avoids being trapped in local optima. They also used the restore operator to amend the non-feasible solutions to ensure the range of solutions. This makes QICA-CARP approximate the most efficient solution. They established the method's effectiveness through experiments and compared the results of their approach with the ones received from the RDG-MAENS and RAM using distinctive test sets. The experimental results indicated that the QICA-CARP outperforms other algorithms regarding convergence rate and the solutions' quality. Notably, the QICA-CARP converges to a better decrease bound faster, illustrating appropriate for solving large-scale CARP.

In [206], Yang et al. proposed a quantum-stimulated ICA (QICA) and a wavelet-packet (WP) based adaptive multi-scale Bandelet (QICAWP-AMB) for image illustration. Multi-level WP decomposition was carried out on the pictures, after which dyadic partition was applied to gain a few squares to carry out Bandeletization. Furthermore, a new price feature independent from the threshold T was proposed to find the optimal path in each square, and the QICA was applied to gain extra accurate geometric flows. Assessments on image coding were taken to compare the efficiency of their proposed model with that of Wavelets, the second generation Bandelet, JPEG2000, SPIHT, GAWP (GA and WP), and ICAPW (ICA and WP) based strategies, and the superiority of QICAWP-AMB was determined in both top signal to noise (PSNR) and edge maintenance index (EPI) at low bit-rate compression.

The quantum-inspired ICA (QICA) is a rising intelligence set of rules. Based on evolutionary game theory and QICA, a quantum-stimulated ICA embedded with the evolutionary game (EGQICA) [207] was proposed to resolve the combination optimization issues. In this paper, the researchers mapped the quantum antibody as locating the top-rated strategy for participants pursuing most software by selecting evolutionary game techniques. Replicator dynamics were used to model the quantum antibody's behaviour, and the reminiscence mechanism was also introduced in this painting. Experimental results showed that the proposed EGQICA constitutes an outstanding diversity and achieves advanced overall performance.

Primarily based on the principles and standards of quality controls, a novel ICA, referred to as the quantum-based ICA (QICA) [208], was proposed to deal with global optimization. In QICA, the antibody is increased and divided into subpopulation groups. The antibodies in a subpopulation organization are represented with the aid of multistate gene quantum bits. The general quantum rotation gate method and the dynamic adjusting angle mechanism are implemented within the antibody's updating to speed up convergence. The quantum not-gate is used to recognize quantum mutation to prevent premature convergences. The proposed quantum recombination performs data transfer between subpopulation groups to enhance the search performance. Theoretical analyses demonstrated that QICA converges towards the global optima. Inside the first part of the experiments, ten unconstrained and thirteen limited benchmark functions were used to check the function of QICA. The results showed

that QICA performs much better than the other advanced GAs regarding the solution quality and computational costs. Li and Jiao [209] proposed a new version of ICA, known as the quantum-stimulated ICA (QICA), which is based totally on the concept and principles of QC and consists of a quantum bit and the superposition of states. Like other EAs, QICA is characterized by illustrating the character, the evaluation characteristic, and the population dynamics. QICA employs a quantum bit, described as the smallest unit of data, for the probabilistic illustration and a quantum bit individual as a string of quantum bits. In QICA, through quantum mutation operator, exclusive use of the current best individual's information is made to perform the following search to speed up the convergence. Information among the subpopulation is exchanged by adopting the quantum crossover operator to improve the population range and keep off prematurity. The researchers executed the proposed QICA to solve the benchmark problems with 30, 100, and 2000 dimensions and large local minima numbers. The results showed that the proposed algorithm could find close-to-optimal solutions with lower computational costs.

3.1.22 22. Social evolution algorithm (SEA)

In [210], a novel Quantum-stimulated SEA (QSEA) was proposed by modelling human bias in individuals' choice for interaction and individuals' capability to emerge from indecisive interactions via looking for a second opinion. The QSEA was experimented on the 0–1 knapsack problem and verified the effect and importance of every human bias in the algorithm's general performance. The results showed that the performance of QSEA is better than or comparable to that of the unique evolutionary algorithmic variations examined. The results imply that the QSE algorithm can be assumed as one of the viable options for solving 0–1 knapsack problems.

3.1.23 23. Shuffled frog leaping algorithm (SFLA)

The collaborative filtering (CF) method's rating score prediction accuracy in social network recommendation systems depends on the nearest associates' extraction and person/assignment similarity calculation. Based on a similar principle to a person/project behaviour, this paper used the most intersection technique to extract the highest-quality neighbour candidate set. It provided a weighted adjusted cosine similarity technique to compute the user/project similarity. Moreover, to optimize the technique's weights, a novel optimization technique called the discrete quantum-stimulated SFL (DQSFL) algorithm [211] was proposed primarily based on the SFLA and QC. ACCORDING TO THE QUANTUM FROG COLONY'S CO-EVOLUTION, the DQSFL algorithm uses quantum movement equations to search for the optimal location. The experiments revealed that the CF recommendation technique, based totally on DQSFL, can efficiently solve the rating information sparse-ness problem in the similarity computation procedure to improve the rating score prediction accuracy and offer a better-endorsed result than the conventional CF algorithms.

Cognitive radio technology has been proposed to enhance spectrum performance by having cognitive radios act as secondary users to access under-utilized frequency bands opportunistically. Spectrum sensing needs to reliably discover signals from certified primary radios to prevent dangerous interference as a critical permitting capability in cognitive radio networks. In [212], researchers proposed a Quantum-inspired SFLA (QSFLA) primar-

ily based on a frog leaping algorithm and examined the function of the QSFLA via a few classical benchmark functions. The simulation results for cognitive radio systems showed that the designed spectrum-sensing algorithm is superior to some previous spectrum-sensing algorithms in terms of the probability of detection, convergence rate, and the accurate convergence value.

3.1.24 24. Space search algorithm (SSA)

Lu and Juang [213] proposed the evolutionary quantum-stimulated SSA (QSSA) to solve numerical optimization issues. Inside the proposed QSSA, the possible solution area is decomposed into state quantum illustration areas. As the search progresses from one generation to the next, the quantum bits evolve progressively to grow the latest areas that render appropriate fitness values. Through the inherent probabilistic mechanism, the QSSA initially behaves as a global search algorithm and progressively evolves into a local search algorithm, yielding excellent stability between exploration and exploitation. An overlapping method was also proposed to prevent an untimely convergence and speed up the overall search speed. The QSSA was implemented on the series numerical optimization problem. The experiments indicated that the results received through the QSSA are pretty competitive compared to those received using the state-of-the-art IPOP-CMA-ES and QEA.

3.1.25 25. Simulated annealing (SA)

The Quantum SA (QSA) [214] is a way to locate the global optima for a COP using quantum fluctuations. QSA may be simulated on a computer using the quantum Monte Carlo (QMC) simulation of the *Ising* model, while sacrificing a big processing time. It has been shown that the processing time of the QMC simulation on a CPU scale is similar to that of QSA on the D-wave 2X QSA (even though the latter is over 108 times quicker than the previous ones). However, massive problems ought to be partitioned into sub-problems and solved one by one, though this reduces the processing speed of the QSA. Because the access to a QSA is also restrained, the acceleration of QA simulations using traditional computers appears a critical topic. If the substantial computational time were reduced, it would be feasible to use QA simulations to solve COPs. An FPGA accelerator has been proposed for the QSA simulations designed using “open computing language”.

Cluster assessment is a common way to segregate data factors into organizations, called clusters. SA is a popular meta-heuristic stimulated using the annealing system used in metallurgy and is beneficial in solving complicated optimization problems. In [215], the QC and SA were used to design a Quantum-stimulated SA method, which may be used to compute the optimum range of clusters for image clustering. Experimental outcomes on some images showed the proposed method's effectiveness in fitness value, precision, convergence time, robustness, and standard error. The paper additionally reports the computation results of a statistical superiority test, referred to as a t-test. Further, an experimental assessment of the classical approach has been provided, demonstrating that the proposed QSA outperforms the others.

3.1.26 26. Tabu search (TS)

Researchers [216] performed a Quantum TS to achieve a quantum combinatorial optimization, arguing that an entanglement-metaheuristic can display the most valuable solutions and speed up the optimization using entangled states. Constructing the best coupling maps showed that the distribution state and outcomes were similar to previous effects in a present teleport circuit. Their goal was to discover which graph state coupling better matches a quantum circuit.

In [217], a novel metaheuristic algorithm called the entanglement-superior quantum-stimulated TS algorithm (Entanglement-QTS) was proposed. The algorithm is based on the quantum-inspired TS (QTS) and the latest quantum entanglement feature. Entanglement-QTS differs from other quantum-stimulated EAs in that its Q-bits have entangled states that may express a high degree of state correlation, rendering the variables more intertwined. Entanglement-QTS can discover a more significant number of solutions, provide stability diversification and intensification, break out several neighbourhoods optimal solutions by using the quantum not-gate, give a boost to the exacerbation impact with local search and the local entanglement search, and control strong-dependence issues and speed up the optimization technique by using entangled states. This study used nine benchmark features to test the search capability of the ultra-modern entanglement-QTS algorithm. The results indicated that the Entanglement-QTS outperforms QTS and other metaheuristic algorithms in locating the global optima and computational performance.

In [218], Kuo et al. proposed a novel algorithm called the Quantum-inspired TS algorithm with Entanglement (QTSWE) that is formed on both the Quantum-stimulated TS (QTS) algorithm and the quantum entanglement trait. QTSWE was used in a deployment problem to determine the minimum number of sensors needed and their locations. This paper analyzed the features of the deployment problem and called it phenomenon dependency. It exploited the concept of quantum entanglement to make the initial positions of sensors and tackle variables' dependency. The QTS was then used to find better solutions iteratively. Also, local search was applied to improve the search capabilities of QTS and avoid being trapped in local optima. The experimental results indicated that QTSWE outperformed others and utilized the few sensors to satisfy the monitoring necessities and topology connectivity. With QTSWE, the function of surveillance system deployment improved further.

In [219], Chou et al. offered a new quantum-stimulated EA called quantum-stimulated TS (QTS). QTS has been formed according to the conventional TS and quantum computation features, such as superposition. QTS has been implemented to solve the 0/1 knapsack problem. Additionally, the experimental results were compared with the original Quantum-inspired EA (QEA) and other heuristic algorithms. The final results showed that QTS operates much better than the others on 0/1 knapsack problems, without premature convergence and higher efficiency.

3.1.27 27. Teaching learning based optimization (TLBO)

The population-based meta-heuristic, known as the TLBO algorithm, created by Rao et al. in 2011 [55], is one of the most effective methods for simulating the traditional classroom-learning process. The Schrodinger equation enhances the Teaching-Learning-Based Optimization algorithm (TLBO) by predicting the likely locations of optimum solutions.

The outcomes showed that by specifying the quantum instructor phase, QTLBO—an abbreviation for Quantum Teaching- Learning- Based Optimization—improved the stability and robustness of the TLBO [220]. To evaluate the effectiveness and quality of QTLBO, two rigid frames for planners and two trusses with strength and serviceability limitations were selected. Comparing the suggested algorithm's findings with those of the standard TLBO algorithm and other literature approaches revealed that QTLBO outperformed the existing TLBO in terms of statistical criteria and enhanced the likelihood of finding a better solution.

A new algorithm was proposed based on the quantum-simulated TLBO (QTLBO) to solve complicated optimization issues [221]. By combining the teaching-learning-based optimization (TLBO) and the QC theory, QTLBO can be developed by the simulated quantum rotation gate function. Then, QTLBO was utilized to resolve linear array pattern synthesis problems. Simulation results indicated that the offered QTLBO strategy could efficiently solve these problems and provide a superior solution to those acquired by previous conventional intelligence optimization strategies. The proposed QTLBO method can search for the optimal global solution for linear array pattern synthesis problems.

3.1.28 28. Vortex search algorithm (VSA)

Li and Zhao [222] introduced an enhanced VSA, a quantum-stimulated type of the latest proposed VSA. The candidate solutions were generated by rotating qubits through a random angle about an axis on the Bloch sphere. The rotation angles of qubits are measured based on the Gaussian random number. The inverse incomplete gamma function was used to modify the variance of the Gaussian distribution. The experimental results of four types of benchmark operations revealed that the proposed QVSA improves the optimization capability of the classical VSA. The statistical results of the CEC 2013 benchmark of the offered QVSA were close to or higher than some of the other state-of-the-art models. Algorithms with fast convergence rates and minor errors are essential for studies seeking high dimensions solutions. Although the single-step frequency time of QVSA is longer, the experimental results indicated that it could acquire the optimal solution in lower iteration steps; therefore, its convergence speed is still quicker than other similar algorithms. Further research will concentrate on improving the computational efficiency of QVSA without decreasing the optimization function.

3.2 Variants of QC

Apart from the early developed metaheuristics, specific metaheuristics developed using QC will be explained in this subsection. Figure (8) shows the QC percentage based on metaheuristic algorithms for the Binary, Chaotic and multi-objective types. As you can see, the multi-objective percentage is more significant than the Binary and Chaotic ones regarding using QC.

A comprehensive summary of the latest QC work has been introduced in Table (6), where publications related to the algorithms in the Binary, Chaotic, and Multi-objective variants employed on different formulations of COPs have been investigated.

A comprehensive summary of the latest QC work has been introduced in Table (7), where publications related to the algorithms in the Binary, Chaotic, and Multi-objective variants employed on different Important Factors.

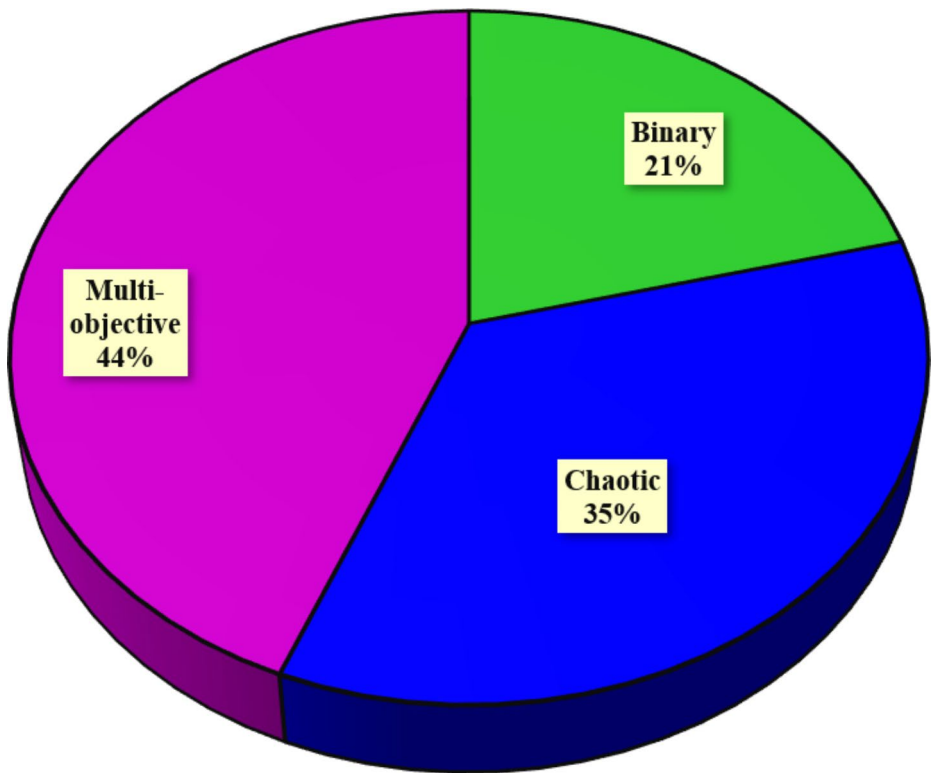


Fig. 8 The percentage of using Quantum-Inspired metaheuristic algorithms for the Binary, Chaotic, and multi-objective algorithms

A comprehensive summary of the latest QC work has been introduced in Table (8), where publications are related to the benefits and weaknesses of algorithms.

3.3 QC-machine learning

Quantum Machine Learning (ML) is an interdisciplinary field that integrates QC with ML. It is a computational communication that utilizes the power of QC to produce quantum versions of ML algorithms [264, 265]. The Quantum Neural Network (QNN) is an Artificial Neural Network (ANN) [266] that is concerned with a system with Quantum Computation. Researchers are working on this area to extend extra efficient algorithms in sample classification or machine learning compared to what is currently available within the capabilities of ANN. Quantum computation extends the computational behavior of machines to exponential potential, using quantum parallelism and the results of intervention and entanglement. Quantum computation is supposed to expedite the training of conventional neural networks and the computation of large data applications, resulting in faster results. A classification of a Decision Tree (DT) is based on a training dataset, which includes observations of objects either acquired empirically or from professionals. The training information in a quantum globe is quantum objects rather than the everyday observations of standard information.

Table 6 Quantum-Inspired Binary, Chaotic, and Multi-objective variants with different problems

Variant	Proposed Algorithm	Problems	Problem domain	publication	Refs
Binary	QGSA-UC	optimization problem	Economic dispatch	Sciencedirect	[223]
	QI-BGWO	optimization problem	Power system optimization	Sciencedirect	[224]
	BQIGSA	Feature selection	Optimization	Springer	[164]
	QOSMRP	The quality of service multi-cast routing problem	Optimization	Inderscience	[225]
	BQIGSA	0–1 knapsack problems, Max-ones and Royal-Road functions	Binary encoded problems	Sciencedirect	[226]
	QBGSA	solving the optimal power quality monitor (PQM) placement problem in power systems	Power quality monitor	Sciencedirect	[227]
	QBGSA	a model of thermal unit commitment problem with wind power integration has been established	Restriction Optimization problem	Sciencedirect	[228]
	QBGSA-KNN	Feature selection	Optimization	Sciencedirect	[229]

The K-Nearest Neighbor (KNN) algorithm, a standard classification method, is highly problematic because it has two significant procedures in computing and searching the similarity time-consuming [266]. This algorithm is one of the simplest and most special methods for data clustering. The basic concept of KNN is that if most of the nearest examples of a point in the functional space belong to an explicit category, it belongs to that category. It has two main procedures: semblance measurement and searching for the nearest samples. Since KNN does not involve any learning and training stages and does not fit the components, categorizing tasks with more and fewer classes is entirely accurate. Quantum KNN (QKNN) uses quantum superposition states to accomplish parallel similitude calculation and uses the minimum quantum search algorithm to accelerate the similitude search process. The Support Vector Machine (SVM) components' choice is impossible without an infinite approach since the choice of SVM components can restrict the improvement of the SVM function. Quantum is used to enhance SVM and its kernel function, as well as the function of SVM. Figure (9) shows the percentage of using QC based on ANN, DT, KNN, and SVM. The ANN percentage is higher in using QC compared to DT, KNN, and SVM.

Table (9) shows the summary of using QC in ML. It has reviewed papers based on objectives and problems.

QC is combined with conventional ANN, giving a novel idea known as QNN that is now explored by different researchers worldwide. The conventional ANN cannot lower learning costs because of its limited computational power. In contrast, the computational power of QNN is higher than its classical counterpart, and one may exploit QNN instead of ANN. QNN takes some benefits of quantum computation, particularly parallel computing traits, and therefore has more robust parallel processing and higher data processing capabilities than the conventional ANN. The benefits of QNN consist of the following:

- Memory capacity is exponential.
- Lower numbers of middle neurons supply higher proficiency.
- Learning occurs rapidly.
- Linearly inseparable problems are used for single-layer network solutions.
- The processing speed is high.
- The catastrophic forgetting is eliminated by not having pattern interference.
- Higher consistency and strength are present.

Obstacles facing the implementation of QNN are as follows:

- Quantum Dependency: The system must keep adhesion until the computation is complete. However, the Quantum system deals with the environment; it is unrealizable to preserve the bonding.
- Connections: The involvement of qubits measures the relationships. This measurement is the primary obstacle against involvement.
- The qubits must form a quantum perceptron to develop an optimal quantum neural network.
- A training algorithm for a QNN must be used instead of exploiting the training algorithms designed for the classical QC.

Table 7 Quantum-Inspired Binary, Chaotic, and Multi-objective with Important Factors

Variant	Proposed Algorithm	Problems	The balance between exploration and exploitation	Convergence speed	Escaping local optimal	Avoid premature optimization	improved search space	update operator	population space	Refs
Binary	QGS-A-UC	optimization problem	✓	✓	-	✓	✓	-	✓	[223]
	QI-BGWO	optimization problem	✓	✓	✓	✓	✓	✓	-	[224]
	BQIGSA	Feature selection	✓	✓	✓	-	-	✓	✓	[164]
	QOSMRP	The quality of service multicast routing problem	✓	✓	✓	-	-	✓	-	[225]
	BQIGSA	0–1 knapsack problems, Max-ones and Royal-Road functions	✓	✓	✓	-	-	✓	-	[226]
	QBGS	solving the optimal power quality monitor (PQM) placement problem in power systems	✓	✓	-	-	✓	✓	-	[227]
	QBGS	a model of thermal unit commitment problem with wind power integration has been established	✓	-	-	✓	-	✓	-	[228]
	QBGS-KNN	Feature selection	✓	-	✓	✓	-	✓	✓	[229]

A classical DT classifier is a learning strategy for discrete-valued classification data. Also, DT can quickly be re-introduced assets of if-then principles. DT categorization can be divided into two sub-problems: (I) create an optimal decision tree classifier with minimum generalization error, and (II) determine the class of the hidden objects with the highest accuracy possible. Since the second problem is trivial for the conventional DT classifiers, most researchers concentrated on the first. The critical role of the first problem is choosing a node-splitting rule. The most fashionable splitting standard consists of information gain, GINI index, chi-squared statistic, and the distance measure. The goal of QC is to carry out parallel data processing and a quick search of the data to render significant advances to the whole field of computer science.

4 Discussion

Numerous quantum-inspired metaheuristic algorithms have been proposed because of their efficiency in global optimization. These algorithms have illustrated promising functions on many problems, and their overall effectiveness across various problems is unlimited. A fundamental EA attempts to make stability between exploration and exploitation. Exploitation and exploration are the opposite aspects of such algorithms. However, the interaction between these terms must be precisely satisfied to make a successful EA. Exploration is kept with QC, which is known to be a successful performer of a global search. Unlike PSO, BCO, FA, IWO, and FPA, each agent uses only the best solution of the best neighbouring agent(s) or the best agent in the population. In QC, each agent uses the best result of all its neighbouring agents. In QC, each iteration attempts to find more appropriate solutions.

4.1 Applications of qc in optimization problems

For example, the high-performance clustering protocol termed quantum clone whale optimization algorithm (WOA) (HPCP-QCWOA) [304] is proposed for sensor node clustering. A novel Adaptive Mutation Quantum-inspired Squirrel Search Algorithm (AM-QSSA) [305] is proposed for global optimization problems. Adaptive Gaussian Quantum behaved PSO algorithm, and Tunicate Swarm algorithm (AGQPSOA-TS) [306] are proposed to optimize the production cost. A quantum entanglement inspired (QEI-GA) [307] is developed to solve the optimization problem. Combining quantum computing with ABC and PSO solves the multi-objective function [308]. An improved quantum-behaved PSO (QPSO) approach has been proposed for multi-objective stochastic project scheduling [309]. Harris Hawk Optimizer (HHO) is extended to deal with dynamic optimization [310].

A dynamic population quantum binary grasshopper optimization algorithm (DQBGOA MR) [332] based on mutual knowledge and rough set theory is suggested for feature selection. A fair balance between exploration and exploitation is obtained through the quantization of grasshopper individuals, which also improves the search scope of feature space. The premature and catastrophe techniques are employed to prevent premature convergence and fall into a local optimum. Furthermore, the rough set and mutual information-based assessment criterion are proposed, which considers the relationship between classes, unselected characteristics, and selected features in addition to the effectiveness of selected features. Twenty UCI datasets are used to assess the DQBGOA MR. According to experimental find-

ings, DQBG OA MR outperforms other swarm intelligence algorithms regarding feature reduction rate, classification accuracy, and stability of results. Figure (10) shows applications of Quantum-Inspired Metaheuristics in different fields.

Thresholding is a simple and effective method used for image processing. The computational time considerably increases in multi-level thresholding as the computation level increases because comprehensive searching adds to the exponential growth of computational intricacy [195]. Therefore, quantum-inspired meta-heuristic methods are utilized to speed up the execution of multi-level thresholding. Frequently, the standard techniques of multi-level thresholding for grayscale images use a binary encoding scheme, where each pixel is represented by bits. Hence, the length of the string is multiplied for higher levels. Quantum-inspired-meta-heuristic algorithms for multi-level thresholding use the notion of qubits, and the real-value encoding scheme is used to assign the active pixel.

Classical optimization methods such as gradient-based optimization methods are incapable of determining the near-global optimal solution due to the following inherent drawbacks:

- The need for continuous and differentiable cost function,
- Converging to the local optimum solution or revisiting the same sub-optimal solution,
- Being unable to search the ample problem space,
- The need for the approximate cost of the initial launch in linear programming,
- Being highly sensitive to starting areas when the number of solution parameters is upgraded, and as a result, the solution space has also increased.

4.2 Advantages and disadvantages

Various metaheuristic algorithms have successfully been executed on optimization problems, but the classical optimization methods have many shortcomings that have been mentioned before. Many researchers have introduced the implications of Quantum-inspired metaheuristics for solving complex real-world engineering optimization problems. It is also observed that QC enhances an algorithm's performance since it is robust when dealing with complex issues. Finding a global optimum in optimization problems can be a troublesome task. QC enhances the exploration virtue of any algorithm by creating ideal points in a search space. In Table (10), the advantages and disadvantages of QC on metaheuristics have been shown.

The difficulty and length of the ACO computations are one of their disadvantages. Although the ACO algorithm outperforms many other optimization algorithms, it is constrained by the requirement for extensive analyses for specific pheromone attributes in each iteration. QACO uses quantum mechanics-based pheromone estimates instead of computations to get over the restriction. Pheromone tracking can be done more precisely while also requiring less computing, thanks to quantum mechanics-based pheromone estimate.

It has been observed that GA is inefficient for the local search in a multi-dimensional search space. Some metaheuristics such as GA, PSO, and DE suffer from immature convergence and are easily trapped in a suboptimal solution. Therefore, to enrich the performance of optimization algorithms in a global search (exploration stage) and local search (exploitation stage), researchers have proposed a substitute technique called QC that is superior to

Table 8 A summarization of the benefits and weakness of QC on variants

Variants	Proposed Algorithm	Problems	Advantages	Weakness	Refs
Binary	QGSA-UC	optimization problem	The high balance between exploration and exploitation, High performance, Low cost	High execution time	[223]
	QI-BGWO	optimization problem	Tradeoff strategy between exploration and exploitation, High Convergence Speed	High iterations	[224]
	BQIGSA	Feature Selection	High Convergence Speed, High Performance, Reducing operating costs	High cost, Without considering all search space	[164]
	QOSMRP	multicast routing problem	High Convergence Speed	High cost	[225]
	BQIGSA	0–1 knapsack problems, Max-ones and Royal-Road functions	The high balance between exploration and exploitation, Low cost	Without considering all search space	[226]
	QBGSA	solving the optimal power quality monitor (PQM) placement problem in power systems	Reducing operating costs, The high balance between exploration and exploitation	High iterations	[227]
	QBGSA	a model of thermal unit commitment problem with wind power integration has been established	The high balance between exploration and exploitation	Without considering all search space	[228]
	QBGSA–K-NN	Feature selection	High Convergence Speed, High Performance	High execution time	[229]

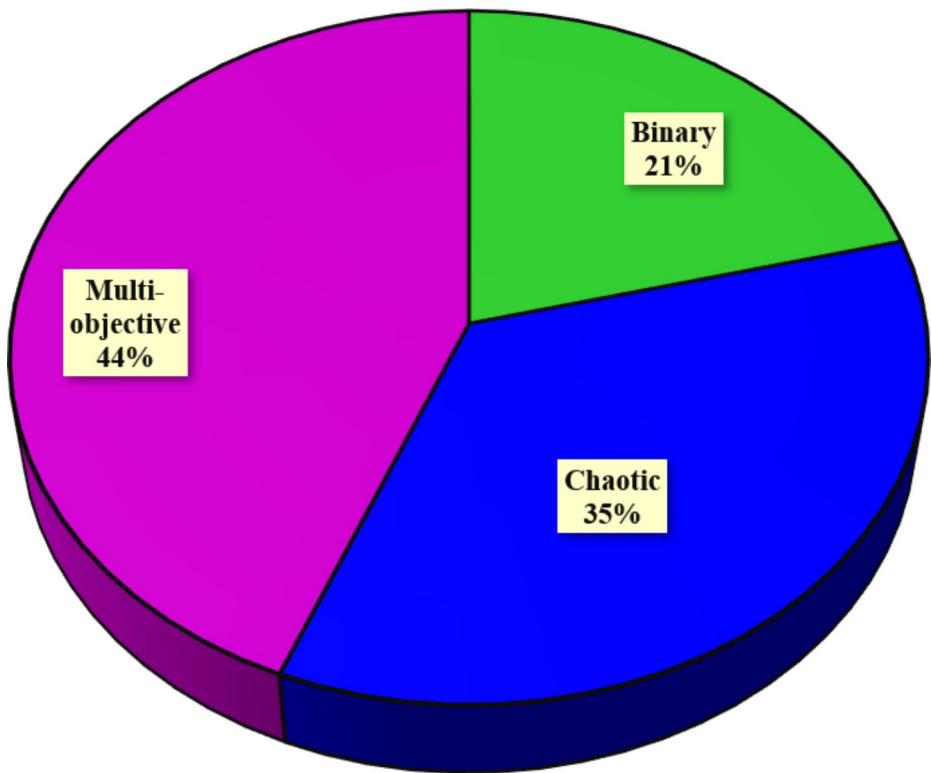


Fig. 9 Percentage of using QC in ML

other methods. QPSO combines the advantages of QC and operators to improve and make the PSO algorithm converge towards the optimal global direction.

The main idea behind QC is to achieve a better approximation for the current candidate solution. In recent literature, the concept of QC has been employed to speed up the convergence rate of optimization problems. The QC is impressive in initialization and in generating new solutions. Metaheuristics are inclined to converge to the local optima. This immature convergence is due to various algorithmic properties, but the main reason is the lack of diversity in a population. Typically, diversity is highest at the onset of the algorithm when all solutions have been randomly generated. As the algorithm develops, diversity generally decreases due to the selection pressure or the sample-generating bias; hence, escaping local optima increases. Using QC samples allows diversity retention and aids in slowing immature convergence and exploring the search space.

4.3 Computational complexity

The development of hybrid QC-based solution approaches for large-scale optimization problems is constrained. The first restriction is to create a hybrid algorithmic framework by combining exact solution approaches with QC-based solutions that use both QC and traditional computers. Another restriction is the development of subproblems that can be

addressed utilizing QC approaches, which have the crucial advantage of computation speed over conventional solution methods. These subproblems' formulation is directly influenced by the characteristics and framework of the main issue. Another research difficulty is defining subproblems that can be solved on a quantum computer.

Last but not least, the resulting subproblems must be manageable enough to be executed on modern quantum systems. For instance, QC-based solution algorithms for operational planning issues can only address small-scale cases and do not yield outcomes that are on par with those attained using the most advanced classical computing techniques. Developing an algorithmic framework that can profit from high-quality solutions obtained using QC approaches is essential to solving such a computational hurdle.

The QC-meta-heuristic algorithms should deliver acceptable solutions for high-dimensional optimization problems while maintaining a low algorithm complexity and a small number of control parameters. The computational complexity for updating the particles and determining the fitness of all particles in the QPSO is $O(2 \times M \times D)$, where M and D are the population size and dimension, respectively [175]. The generation of n new particles ($n < M$), with an $O(n \times D)$ runtime and an $O(n \times D)$ complexity for computing fitness, is necessary to expand the population's diversity. A runtime of $O(M \times D)$ is required to update the personal best and overall best positions. Based on the evaluations above, the suggested method has an overall computational complexity of $O(P \times (M + n) \times D)$, where P is the total number of generations.

In contrast, the computational complexity of PSO is $O(P \times M \times D)$. The simulation results revealed that the exploration and exploitation of QPSO have significantly developed. In these practical applications, the QPSO also obtained notable results, illustrating that the qualified algorithm is a valuable auxiliary appliance in solving complex optimization problems. Algorithm QPSO has a shorter running time than PSO, ACO, and GWO. In the time-consuming PSO method, three strategies (search, swarm, and follow) are calculated and compared in each iteration. The accuracy and size of the feature subset are higher with QPSO.

The number of solutions is denoted by n , the size of agents is denoted by d , and the number of iterations is denoted by t . Also, the population initialization is executed p times. It takes $O(n \times d)$ time to convert qubit grasshoppers to binary grasshoppers. Each grasshopper's position must be updated, which requires $O(d)$ time complexity. In method DQBGOAMR [311], a dynamic population strategy is employed to obtain a better solution that consumes time complexity $O(n)$. As a result, the overall time complexity of DQBGOAMR is $O(n \times d + p) + O(t \times n \times d) + O(t \times n)$, which can be represented as $O(n \times d + p + t \times n \times (d + 1))$.

4.4 Statistical tools

A set of evaluation criteria evaluates each new algorithm. Various criteria have been used to evaluate and compare QC models. The most important evaluation criteria are as follows:

- QC has higher detection precision. Precision of diagnosis is related to classification, feature selection, prediction, estimation and clustering.
- QC has a higher accuracy percentage. Percentage belongs to the classification, clustering, data training, validation and evaluation of data in the knowledge base.

- QC has a lower error rate. QC reduces errors in disease diagnosis, image segmentation, and continuous and discrete optimization problems.
- QC is run based on iteration. QC has achieved the optimal solution with less execution rate in comparison with conventional algorithms.

5 Conclusion and future works

This work revised and classified metaheuristic methods relating to the QC technique. Over 180 research papers were studied and analyzed in this review to make a robust conclusion for the researchers interested in Quantum-Inspired metaheuristics. In detail, this study summarized the references published between 1996 and the first half of 2019. Many optimization methods such as QC have been proposed in the last three decades to find acceptable solutions for various optimization problems, where mathematical gradient-based methods cannot be used. In particular, in evolutionary algorithms such as DE, PSO, HSA, Simulated Annealing, and ACO, QC has been implemented to improve metaheuristic search processes. Also, these methods have been implemented to solve COPs and MOOPs and continuous and discrete optimization problems. Different problems in the real world involve varying exploration and exploitation capabilities, which control parameters can manage. Many works should be devoted to adapting the QC to the parameters of metaheuristics in different applications and investigating each parameter's effects on the convergence rate and getting trapped into local optima.

This paper studied the formational aspects of all the significant metaheuristics inspired by QC. It is hoped that the content presented will be helpful to researchers, and they will be able to use QC and metaheuristic algorithms in their studies to solve optimization problems. This paper could also guide researchers currently working or will study in this field by leading them towards how the Quantum-Inspired metaheuristics can be employed to solve problems, point out their strengths and shortcomings, and prove their effectiveness. Thus, significant problems can still be solved using Quantum-Inspired metaheuristics.

In reviewing previous studies, it was found that some papers have gaps that should be considered in the following manner:

- The objective functions are formulated concerning fulfilling the constraints. However, all the optimization problems' limitations must be considered to achieve the optimal solution for global convergence. However, the maximum number of constraints must be considered to make the complex problems more accurate.
- QC has been used in numerous optimization problems; however, they are different in terms of convergence rate, complexity, search capability, and initial parameters. Thus, developing meta-heuristic algorithms based on QC requires extended time and execution complexity.
- The performance of meta-heuristic algorithms can be further enhanced by integrating advanced operators. Nevertheless, there are concerns with the deployment of QC in meta-heuristic algorithms. QC must be in the right place for the optimization operators (such as crossover and mutation in GA), be compatible with the meta-heuristic

algorithms, and not reduce the algorithm's speed. Hence, further studies should be conducted on integrating optimization problems with QC.

The information provided in this review can be vital in improving optimization problems and exploration and exploitation in future research applications.

The study contributions were organized, illustrated, and summarized in tabular and graphic formats to track the review content easily. QC has been applied to combinatorial, dynamic, large-scale, constrained, multi-objective optimization and artificial neural networks. QC applications include image processing, clustering, network, electrical engineering problems, mathematical equation problems, etc. QC has proven helpful for the optimization and engineering community and researchers currently working or working in these areas. The purpose of this review was to present a relatively comprehensive list of all the Quantum-inspired Metaheuristic Algorithms to inspire further research. The main limitations of this paper include the lack of consideration of parameters in optimization algorithms, the accuracy of QC-based metaheuristic algorithms by various experiments, the lack of computational time on the QC-based model and the lack of review of iterations in standard and quantum models. Future works will be needed to study optimization problems by integrating QC equations in metaheuristic algorithms to strengthen solutions. We will study multi-strategy and use different methods to model and optimize the overall performance of QC.

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Springer Nature or its licensor holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.**Fig. 10** Applications of Quantum-Inspired Metaheuristics in different fie
l d s

Table 10 Advantages and disadvantages of QC on metaheuristics

Advantages	Disadvantages
} fast convergence } convergence with a lower error average } improving the convergence speed } maintaining equivalence between exploration and exploitation } incrementing the global searchability } the probability update rule } incremental learning based on rotation gate } greater exploration of the search area } making a robust solution for optimization problems } making robust solutions for optimization problems based on the state of different quantum } The jumping rate for ith agent in the population is produced based on a quantum distribution	x The relative increase in computational time due to testing different states x Increasing iterations - - - - - - - - - - -

